

Identifying Variation, Facility Size, and Capacity in the UST Replacement Model

Diagnostic report on what the structural model uses and what it currently ignores

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1 Setup

This report documents four things:

1. **Where the dynamic replacement model gets its identifying variation** — visualizing the cross-cell joint distribution of premium P and hazard h , the time-series shifts in P , and the empirical action heterogeneity.
2. **The size dimension (`n_tanks_active`)** — empirical evidence that facility size, currently averaged away by the model, is a meaningful driver of action probabilities.
3. **The capacity dimension (`total_capacity`)** — analogous evidence for total tank capacity in gallons, distinct from tank count.
4. **Facility-portfolio behavior around closure events** (Section 5) — what does a tank closure actually mean for the facility? Classifying every closure event by the facility's subsequent install behavior, and mapping those facility-level realities to current and proposed

model actions.

Sample throughout: the canonical observed DCM panel (TX 2006+, controls 1999+; $N = 2,282,735$ facility-year observations across 139,493 unique facilities). The fit referenced for model-implied quantities is the 6-parameter universal- γ stayFE-profile specification with SD and MO dropped (Model_Replacement_6paramFE_profile_observed_dropSDMO.rds from Ticket 005).

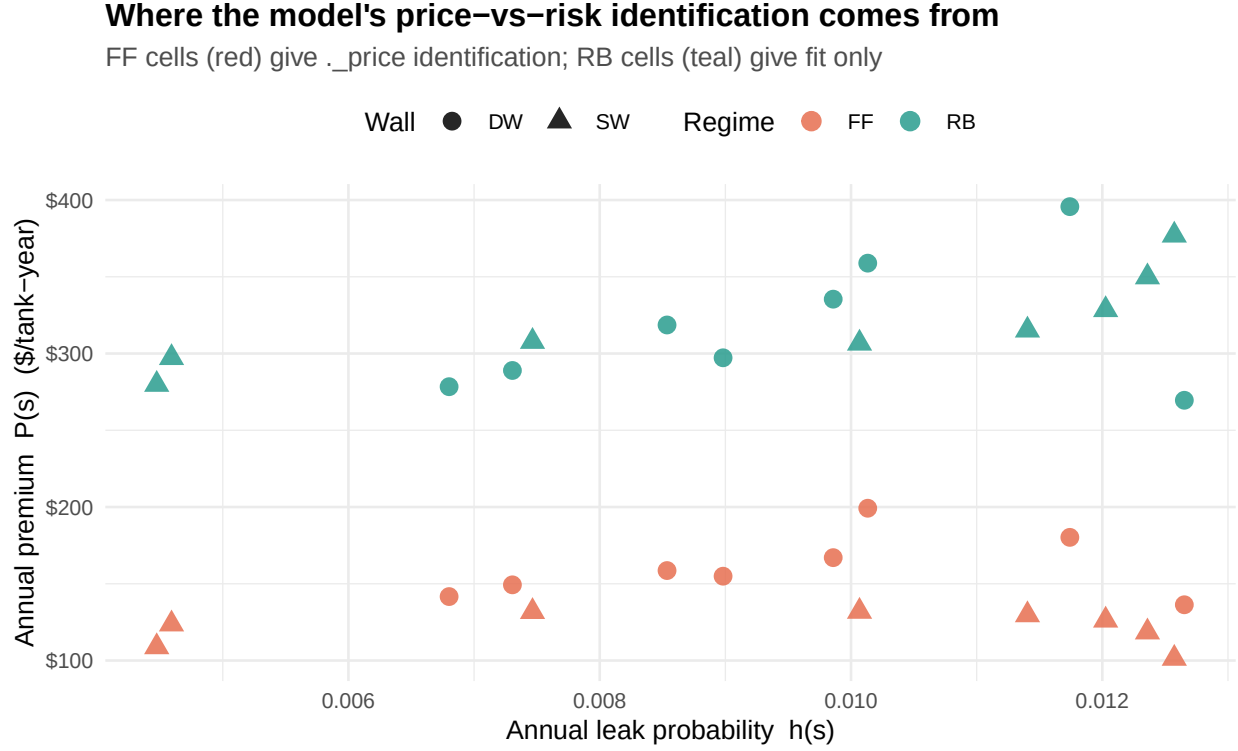
The takeaway upfront: **the joint (P, h) variation cleanly identifies the universal- γ structural parameters**, but **facility size and capacity capture systematic behavioral heterogeneity that the current 32-state model averages away**, and **the model’s action definitions (Exit vs Replace vs Maintain) conflate genuinely different facility-portfolio behaviors**. Section 4 quantifies the action-definition issue: how much of the current “Exit” data is actually a partial-shrinkage event where the facility kept operating, and how much is a true facility-level walk-away.

2 Identifying variation in the structural model

2.1 Premium $\times$ hazard cell-level scatter

The structural identification of $(\gamma_{\text{price}}, \gamma_{\text{risk}})$ requires that P and h vary *independently* across some subset of the data. Figure 1 plots the 32 state cells of the model in (P, h) space, colored by regime. **Under FF**, P is set administratively and is uncorrelated with h within the regime — that is the source of clean γ_{price} identification. **Under RB**, P is risk-priced (covaries with h); the RB cells provide fit but cannot separately identify γ_{price} from γ_{risk} .

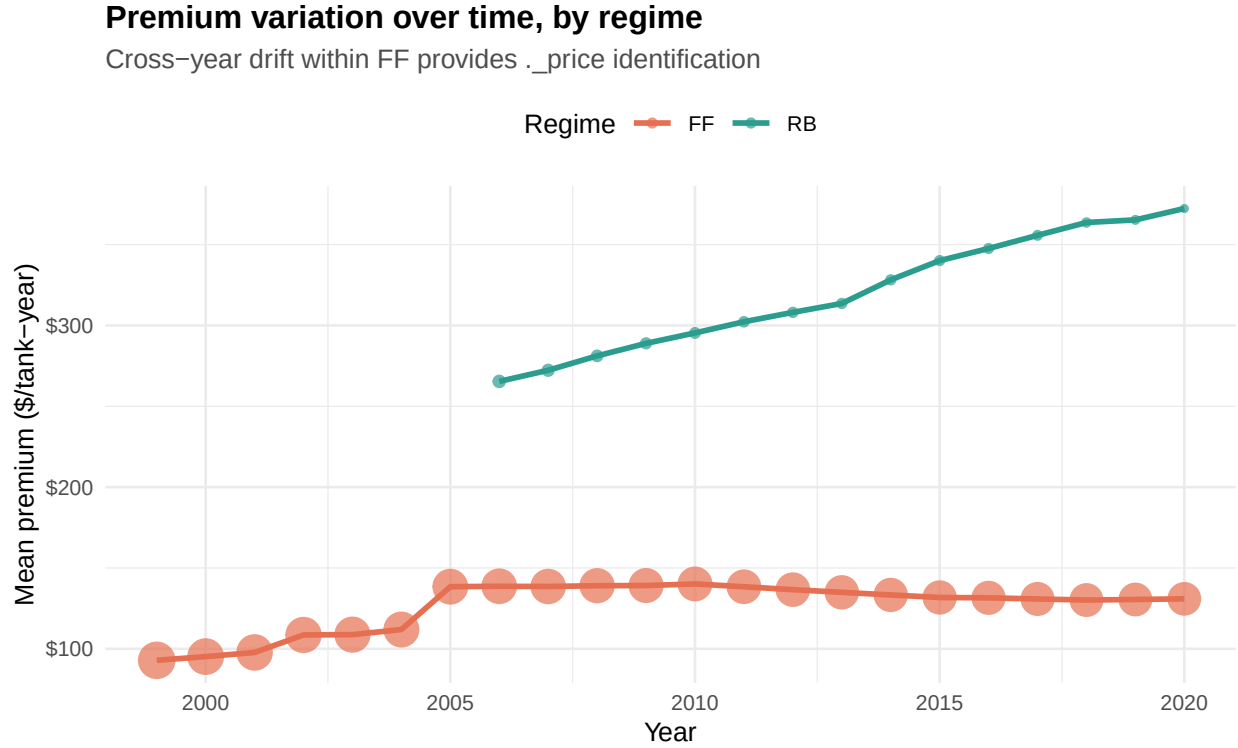
Figure 1: Cell-level joint distribution of annual premium and hazard probability across the 32 state cells of the structural model. Each point is one (age $\times$ wall $\times$ regime) cell. FF cells span a range of hazards at roughly constant premium; RB cells show premium covarying with hazard.



2.2 Cross-state, cross-year premium variation within FF

Within the FF regime, premium variation across states and over time provides additional identifying power for γ_{price} . Figure 2 shows year-by-year mean premium by regime in the DCM sample. The pre-2006 TX (extended sample, included for context) had a different premium structure than post-2006; control-state FF premiums also drift over the sample window.

Figure 2: Mean annual premium per tank by regime and year. TX shows the FF-to-RB transition at the 2006 reform; control states have continuous FF premium drift.



2.3 Empirical action shares across the 32 state cells

Table 1 summarizes the empirical action shares across the 32 cells. This is the variation the structural likelihood is trying to fit; the fit figures in `Dynamic_Model_Fit_Report.qmd` show the model-implied overlay.

Table 1: Empirical action shares by (wall $\times$ regime $\times$ age bin) cell, with cell sample size n and dollar premium / hazard-loss values.

Cell	Wall	Reg	Age	N obs	Premium USD per yr	Haz loss USD per yr	Maintain pct	Exit pct	Replace pct
1	SW	FF	1	46,709	109	2925	98.33	1.46	0.201
2	SW	FF	2	97,999	124	2991	98.52	1.13	0.350
3	SW	FF	3	180,163	132	3602	97.49	2.17	0.340
4	SW	FF	4	240,344	132	4784	97.15	2.58	0.271
5	SW	FF	5	255,278	130	5307	97.19	2.63	0.174
6	SW	FF	6	234,030	126	4954	97.69	2.20	0.113
7	SW	FF	7	191,377	118	4351	97.95	1.95	0.092
8	SW	FF	8	410,728	101	3780	98.52	1.45	0.032
9	DW	FF	1	64,621	136	4992	99.90	0.00	0.099
10	DW	FF	2	77,103	142	2280	99.36	0.00	0.636
11	DW	FF	3	77,966	149	1948	99.08	0.00	0.920
12	DW	FF	4	62,241	155	2043	98.93	0.00	1.067
13	DW	FF	5	41,520	159	1454	99.31	0.00	0.686
14	DW	FF	6	20,407	167	1502	99.74	0.00	0.265
15	DW	FF	7	4,872	199	2165	99.47	0.00	0.534

(continued)

Cell	Wall	Reg	Age	N obs	Premium USD per yr	Haz loss USD per yr	Maintain pct	Exit pct	Replace pct
16	DW	FF	8	2,867	180	3733	99.62	0.00	0.384
17	SW	RB	1	3,350	280	2925	99.37	0.54	0.090
18	SW	RB	2	9,999	297	2991	98.81	1.12	0.070
19	SW	RB	3	19,093	308	3602	98.73	1.04	0.225
20	SW	RB	4	26,212	307	4784	97.75	2.06	0.198
21	SW	RB	5	36,841	315	5307	97.27	2.58	0.149
22	SW	RB	6	37,789	328	4954	97.32	2.48	0.198
23	SW	RB	7	31,888	350	4351	96.85	2.92	0.223
24	SW	RB	8	30,722	377	3780	95.79	4.01	0.205
25	DW	RB	1	18,813	270	4992	99.95	0.00	0.053
26	DW	RB	2	16,557	278	2280	99.43	0.00	0.568
27	DW	RB	3	14,499	289	1948	99.26	0.00	0.738
28	DW	RB	4	13,106	297	2043	98.02	0.00	1.976
29	DW	RB	5	8,677	319	1454	97.49	0.00	2.512
30	DW	RB	6	4,218	335	1502	98.29	0.00	1.707
31	DW	RB	7	1,770	359	2165	98.08	0.00	1.921
32	DW	RB	8	976	396	3733	98.16	0.00	1.844

3 The size dimension: number of tanks per facility

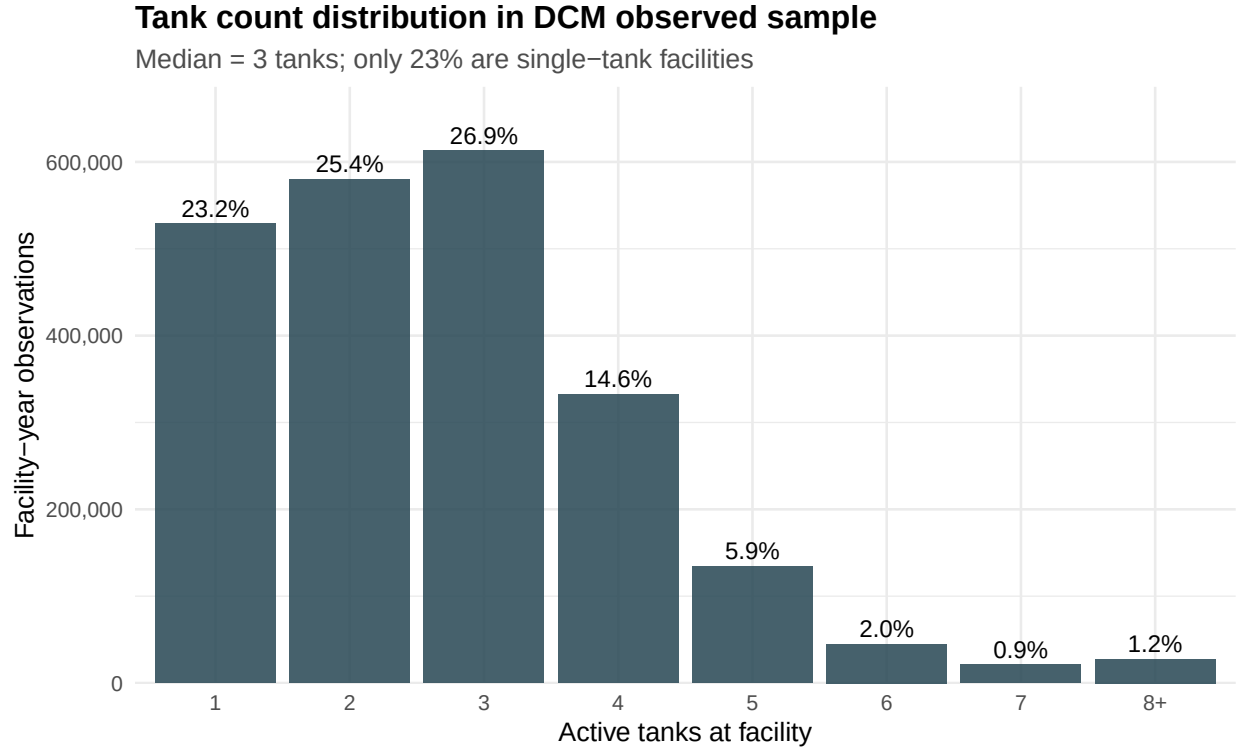
3.1 Distribution

The DCM sample has substantial spread in facility size. Only **23%** of facility-years are single-tank; the median facility-year has **3 tanks**; 24% have **4 or more**.

Table 2: Empirical action shares by (size $\times$ wall), pooled across regime and age. Single-tank DW facilities show literally zero Replace events; 4+ tank DW facilities replace at roughly 4 percent per year.

Wall	Size	N obs	Maintain pct	Exit pct	Replace pct
DW	1	106,328	100.000	0.000	0.000
DW	2	142,131	99.888	0.000	0.112
DW	3	111,908	99.845	0.000	0.155
DW	4+	69,846	96.004	0.000	3.996
SW	1	422,684	97.015	2.958	0.027
SW	2	438,082	97.757	2.173	0.070
SW	3	501,008	98.026	1.850	0.124
SW	4+	490,748	98.138	1.446	0.417

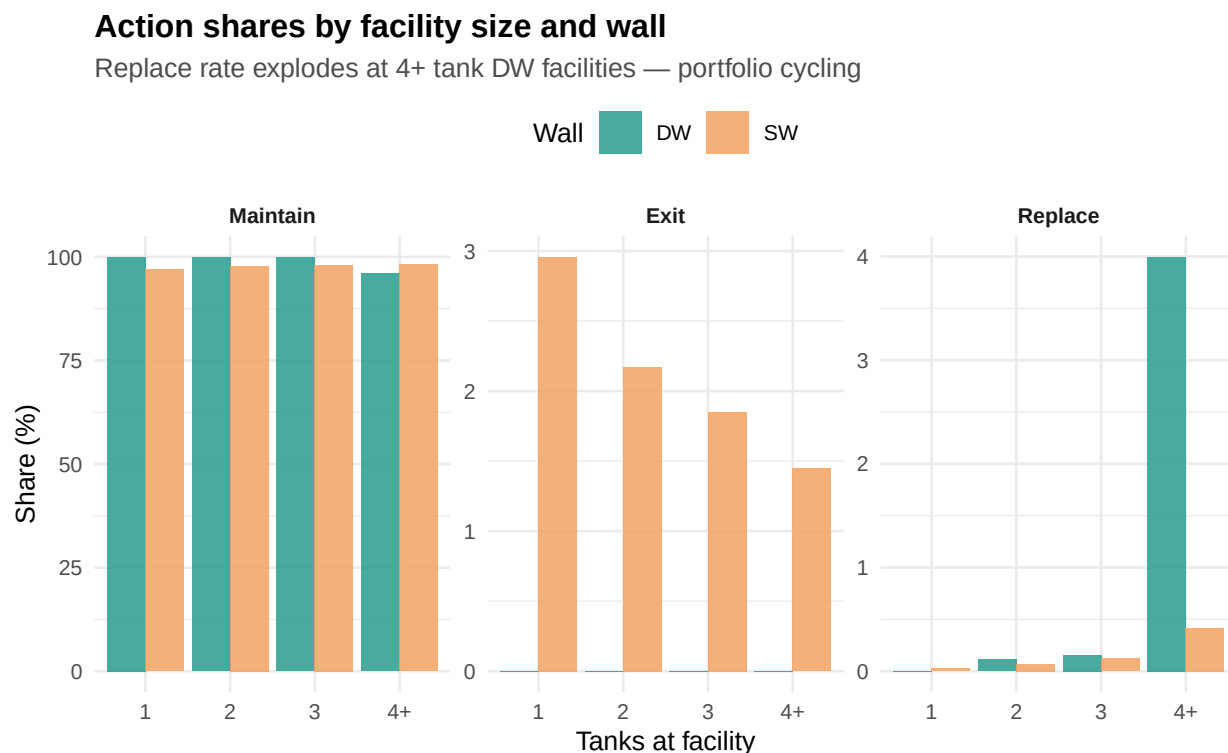
Figure 3: Distribution of active tank count per facility-year in the DCM observed sample. Bars truncated at 8+; about 1.6% of facility-years have ≥ 8 tanks.



3.2 Action shares by (size $\times$ wall)

Table 2 and Figure 4 together show the striking size-action relationship. The pattern is most dramatic at the **Replace** margin for double-wall tanks: DW replace rate jumps from **0% at single-tank facilities to ~4% at 4+ tank facilities**. This is almost certainly the portfolio dynamic — large multi-tank facilities cycle individual DW tanks at a high rate that the model attributes to “Replace” but is really within-facility capital refresh.

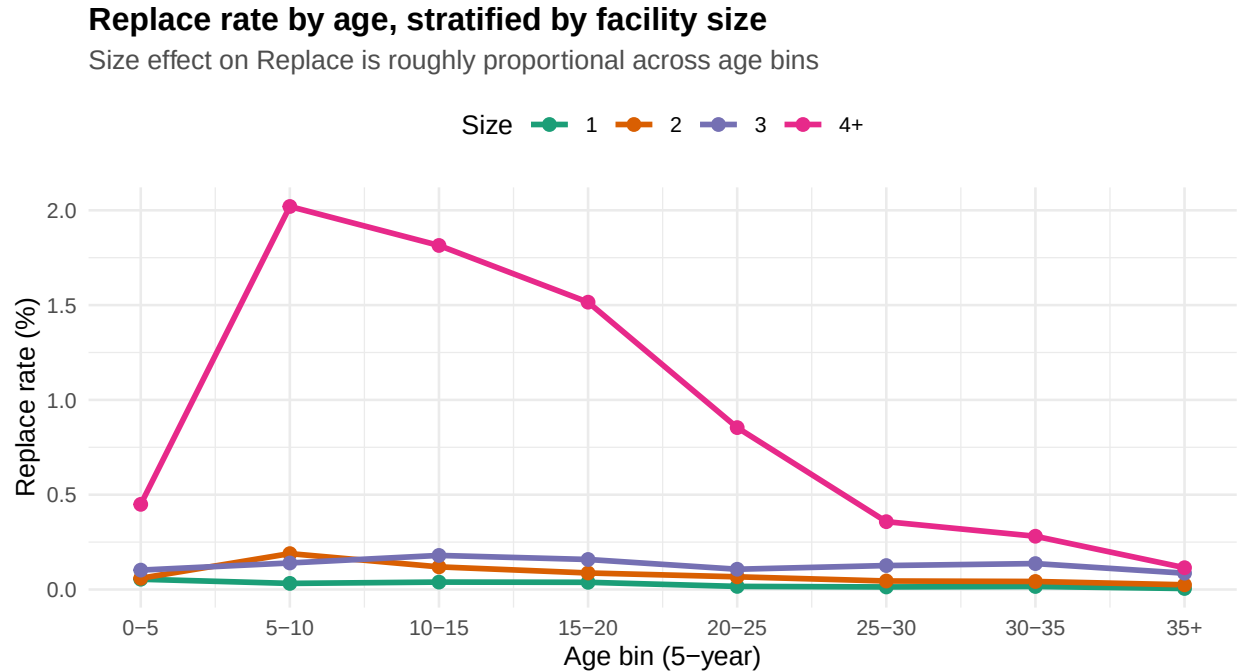
Figure 4: Empirical action shares by facility size and wall. Replace rate (right panel) grows steeply with size, especially for DW; this is the portfolio-refresh signature.



3.3 Action shares by (size $\times$ age)

Figure 5 shows whether the size pattern is stable across age bins or interacts with age. The Replace excess in 4+ tank facilities is visible across all ages; it is not driven by any single age bin.

Figure 5: Replace rate by age bin, separately for each facility size class. The 4+ tank line sits above the others at every age.

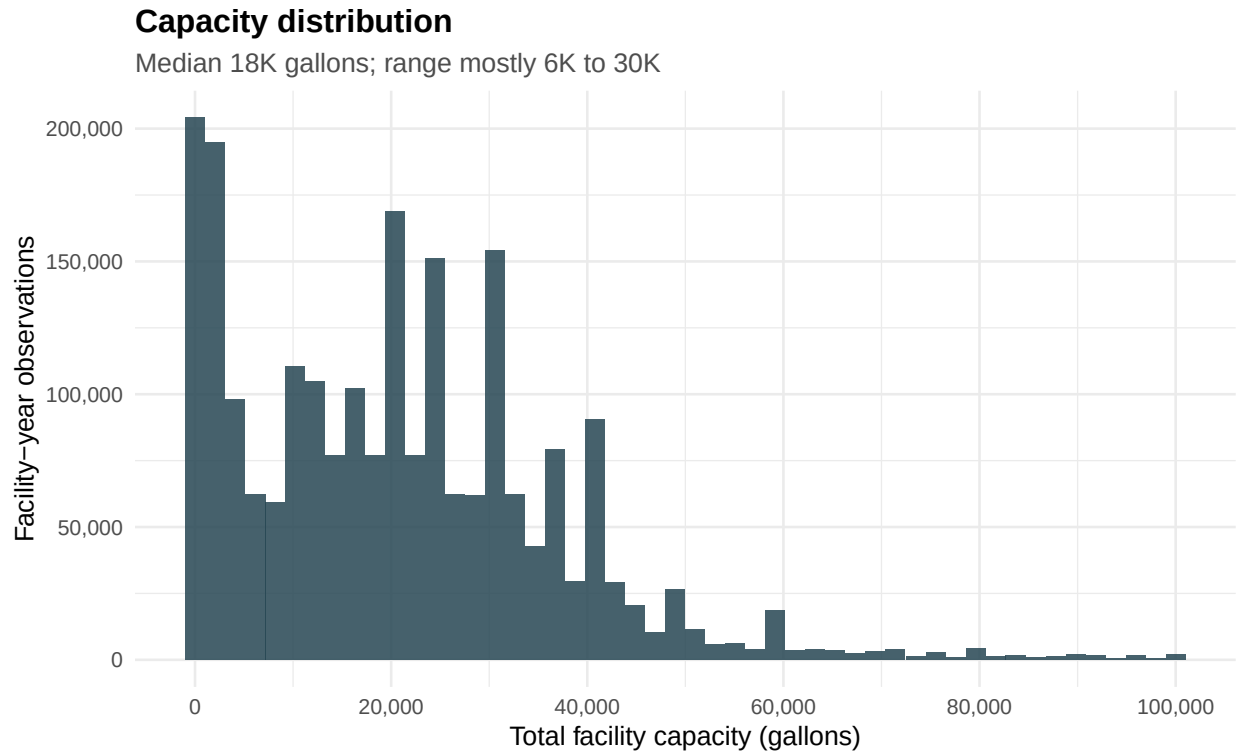


4 The capacity dimension: total gallons at facility

4.1 Distribution

Total facility capacity is correlated with tank count but adds orthogonal information. The DCM sample's 25th/50th/75th percentiles of total capacity are **6K / 18K / 30K gallons** — a typical small station has one or two 12K tanks; a high-volume station has four 20K tanks (~80K total).

Figure 6: Distribution of total facility capacity in gallons. Truncated at 100K for readability; 1.9% of facility-years exceed 100K total gallons.



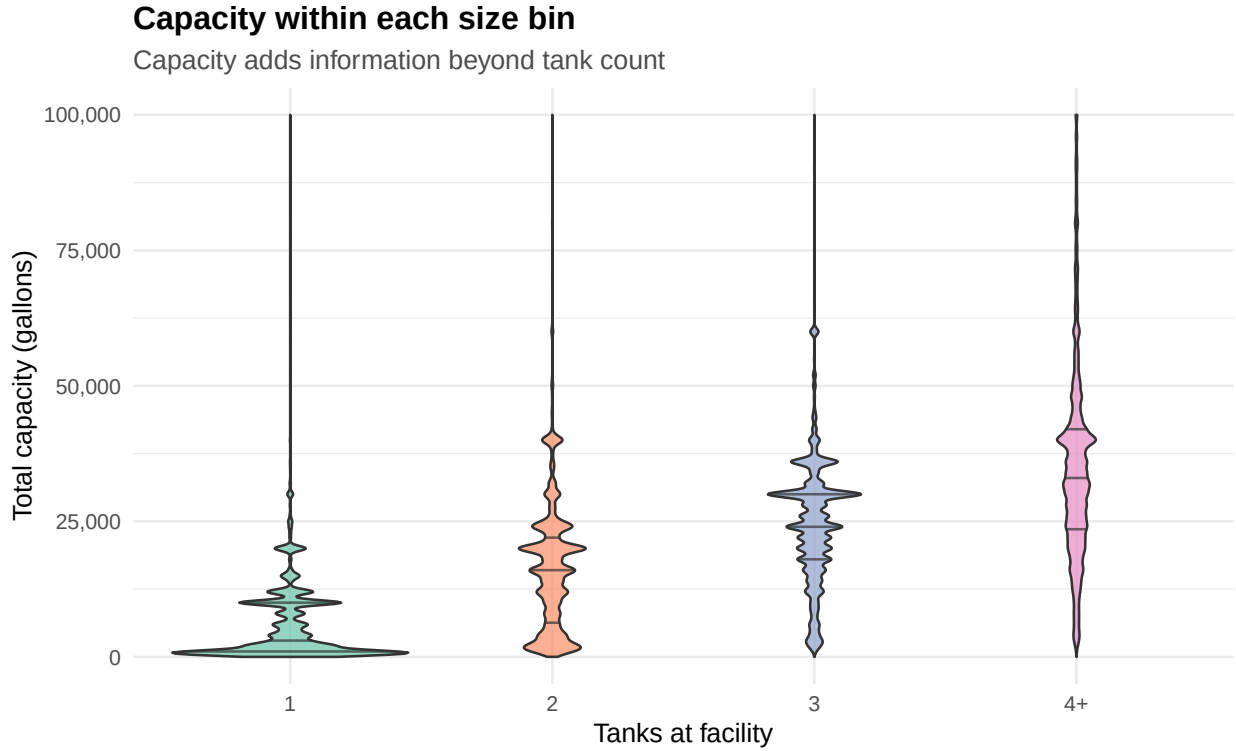
4.2 Capacity vs tank count

Capacity is partly mechanical (more tanks $\rightarrow$ more gallons) but also captures per-tank size. Figure 7 shows the joint distribution: within each tank-count bin there is substantial spread in total capacity, meaning capacity adds orthogonal information beyond size.

Table 3: Empirical action shares by total-capacity bin, pooled across wall, regime, and age. Replace rate grows with capacity (consistent with the size pattern but distinguishable).

Capacity bin	N obs	Maintain pct	Exit pct	Replace pct
<10K	638,005	97.336	2.596	0.069
10-30K	990,436	98.239	1.620	0.141
30-60K	572,997	98.619	0.936	0.444
60K+	81,297	97.232	0.519	2.249

Figure 7: Distribution of total capacity within each tank-count bin. Each violin plot shows the spread of total gallons for facilities with that many tanks; horizontal lines mark median, 25th, 75th. Capacity is positively related to tank count but not redundant — there is meaningful within-bin spread.



4.3 Action shares by capacity bin

Table 3 mirrors the size analysis but slices by total capacity. The same Replace-rate growth shows up, with the largest capacity bin (60K+ gallons) having a Replace rate of **2.25 percent** — roughly 33 times higher than the smallest bin (0.07 percent).

4.4 Disentangling size from capacity

If we believe capacity and tank count are both proxies for “facility scale”, does either subsume the other? Table 4 cross-tabulates the two and shows Replace rate in each (size, capacity) cell. The pattern that emerges: **tank count dominates capacity in this sample**. Almost every cell at

Table 4: Replace rate (%) by (size $\times$ capacity) cell. Row $\times$ column shows both dimensions independently shift the Replace rate.

Tanks	Total capacity (gallons)			
	<10K	10-30K	30-60K	60K+
1	0.016	0.035	0.016	0.000
2	0.104	0.068	0.084	0.052
3	0.165	0.118	0.138	0.093
4+	0.306	0.398	0.776	2.886

4+ tanks (Replace rates 0.31–2.89 percent) exceeds every cell at 1–3 tanks (Replace rates 0.00–0.16 percent), regardless of capacity bin. Within 4+ tank facilities, capacity also matters (0.31 percent at <10K vs 2.89 percent at 60K+), so the two dimensions are not redundant – but tank count is the first-order driver of Replace behavior.

5 What does a tank closure actually mean at the facility level?

In the structural model, a “Replace” event means the agent paid K to install a new (DW) tank; an “Exit” event means the agent took the outside option and walked away. At the facility level, what actually happens when a tank closes? This section classifies every closure event in the DCM sample by the facility’s subsequent install behavior and shows how that classification maps onto the current and proposed model actions.

5.1 Setup: closure-event taxonomy

For each facility-year with at least one tank closure, we look ahead at the same facility’s panel history and classify the event by what follows. The taxonomy:

- **Same-year swap** — the facility installs at least one new tank in the same year as the closure ($n_{\text{install}}[t] > 0$ and $n_{\text{close}}[t] > 0$). The cleanest possible “retrofit cycle”: one tank out, one in, same year.
- **Delayed retrofit (1-5 yrs)** — no install in the closure year, but at least one install at the same facility within 5 years.
- **Long-gap retrofit (> 5 yrs)** — install eventually, but more than 5 years after the closure.
- **Partial shrinkage** — facility closed some tanks, kept others, but *never installs again* in the panel. Facility still operating with fewer tanks.
- **True exit** — all tanks closed this year ($n_{\text{eoy}}[t] = 0$) AND no future install at this facility anywhere in the panel.

This is the user’s plain-English mapping: - Replace = closure followed by future install at the same facility (any gap length) - Exit = all-tanks-gone with no future install ever (truly leaving) - Maintain = facility kept operating, including partial-closure cases where some tanks remain (the facility didn’t make a structural-model decision; just adjusted its tank portfolio)

Note on counts vs. T005. The earlier “T005 has 6,210 Replace events” diagnostic used 04b’s

Table 5: Distribution of facility-year closure events by what follows at the same facility. Each row is a category of facility-level outcome conditional on at least one tank closing that year.

Outcome	N closure events	Pct of closures
Same-year swap	6,823	12.9
Retrofit 1-5 yrs	1,191	2.3
Retrofit >5 yrs	327	0.6
Partial shrinkage	6,467	12.3
True exit	37,960	71.9

I_replace flag, which is built on 02b’s tank-level `is_replacement = (fac_max_install_date > tank_closed_date)` condition. That condition requires the next install to occur strictly *after* the tank’s closure date. The “Same-year swap” category in the taxonomy here lumps in all closures and installs occurring in the same calendar year regardless of chronological order, so the taxonomy’s retrofit total (8,341 closures = same-year + 1-5y + >5y) is about 2,100 higher than T005’s Replace count. The gap is closures where the facility installed *earlier* in the same year (then closed a different tank later) – which T005 treats as Maintain at the facility-year level but the taxonomy treats as a swap. Both approximations are reasonable at year-level granularity; the user’s plain-English definition is closer to T005’s chronological rule.

5.2 Distribution of closure-event types

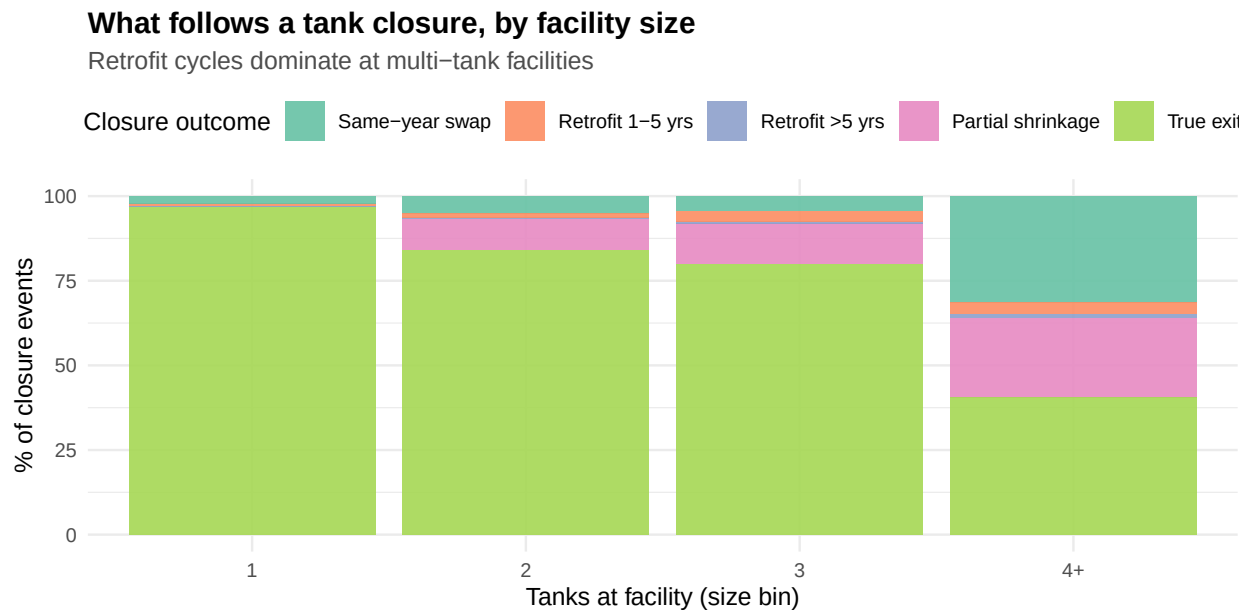
Table 5 and Figure 8 show the overall composition of closure events under this taxonomy and how it varies by facility size. **True exits dominate** at 72 percent of all closure events, followed by retrofit cycles at 16 percent (same-year swap 13 percent plus delayed retrofit 3 percent) and partial shrinkage at 12 percent.

The size-stratified composition (Figure 8) tells a sharper story:

- **Single-tank facilities (1 tank):** 97 percent of closures are true exits. Almost no retrofit activity. By construction these can’t partially shrink.
- **Small multi-tank facilities (2–3 tanks):** still dominated by true exits (80–84 percent), with partial shrinkage at 9–12 percent and retrofit cycles at only 5–8 percent.
- **Large multi-tank facilities (4+ tanks):** the pattern flips – true exits drop to 41 percent, retrofit cycles rise to 36 percent (mostly same-year swaps at 32 percent), and partial shrinkage takes 23 percent.

The 4+ tank facilities are the *only* group where retrofit cycles are a majority-ish share of closure events. For the other ~75 percent of the sample, closure events overwhelmingly mean “facility leaves the industry.”

Figure 8: Composition of closure events by facility size. Single-tank facilities are roughly half retrofits and half true exits. Larger facilities are dominated by same-year swaps and retrofit cycles — the portfolio refresh signature.



5.3 Closure-to-install gap distribution

For closure events that ARE eventually replaced, Figure 9 shows the distribution of years until the next install at the same facility. **Most retrofits happen within the same year as the closure**, with secondary peaks at 1-2 years and a long tail.

Table 6: Composition of closure-and-install events. Rows: type of tank(s) closed. Columns ‘SW’, ‘DW’, ‘Both’ = wall type of same-year install at the facility. Column ‘Within 5y’ = no same-year install but install within 5 years. Column ‘Never’ = no install at this facility for the rest of the panel. Cells are counts of closure events.

Closed	SW	DW	Both	Within 5y	Never
SW closed	1786	3475	48	973	36111
DW closed	11	964	3	148	4527
Both closed	4	267	3	27	473

Figure 9: Distribution of years from closure to next install at the same facility, among closure events that are eventually replaced. Same-year swaps (gap = 0) dominate; the tail extends to multi-year delays.



5.4 Composition shifts around closure events

For each closure event, did the facility’s wall composition shift? Table 6 cross-tabulates what was closed against what was installed (in the same year or within 5 years). This reveals the *directional* tank dynamics: SW → DW (retrofit upgrade), DW → DW (capital refresh), SW → none (downsize), etc.

The diagonal entries (SW closed → SW installed; DW closed → DW installed) show direct tank-for-tank substitution. **Off-diagonal cells where the wall type changes — SW closed → DW installed — are the structural retrofits the model conceptually targets**, but they’re a small share of the table.

5.5 Mapping facility-level reality to model actions

The whole point of the taxonomy: how do these facility-level events map to model actions, both under the current T005 definition and under the proposed clean definition?

Table 7: Mapping of facility-level closure outcomes to model actions. T005 column shows the current (04b) classification; Clean column shows the proposed (T006) classification per the user’s plain-English definitions. Mismatches are highlighted.

Outcome	N events	Pct of closures	T005 action	T006 action
Same-year swap	6,823	12.9	Replace	Replace
Retrofit 1-5 yrs	1,191	2.3	Replace	Replace
Retrofit >5 yrs	327	0.6	Replace	Replace
Partial shrinkage	6,467	12.3	Exit (MISMATCH)	Maintain
True exit	37,960	71.9	Exit	Exit

Table 8: Per-tank closure rate (any tank in the facility closed that year) vs full-facility-closure rate, by facility size. Larger facilities have much higher per-tank rates relative to full-facility-closure rates — reflecting that individual-tank events at multi-tank facilities are common.

Size	N obs	Any closure pct	Full closure pct	Any to Full ratio
1	531,814	2.375	2.375	1.0
2	585,216	1.908	1.665	1.1
3	618,146	1.875	1.571	1.2
4+	563,643	3.084	1.357	2.3

The single substantive mismatch is the Partial shrinkage row: a facility closed some tanks but kept operating and never installs again. T005 calls this Exit; T006 calls it Maintain. The numbers in Table 7 quantify how much of the current “Exit” data is actually this case.

Two implications fall out of this:

1. **T005’s Exit margin is contaminated** by partial-shrinkage events. The estimated $\hat{\kappa}$ partly reflects “value of partial downsizing,” not “value of full facility exit.”
2. **T005’s Replace data is mostly clean.** The taxonomy bins “Same-year swap,” “Retrofit 1-5 yrs,” and “Retrofit >5 yrs” all correctly map to Replace under both T005 and T006. The Replace total is the same in both classifications; the issue with the *interpretation* of Replace (per Section 4.4 of this report) is the structural one — `replacement_closure_year` captures any tank-replacement, including capital refresh at multi-tank DW facilities, which is conceptually distinct from SW→DW retrofit.

5.6 Per-tank vs facility closure rates

For completeness, Table 8 shows the rate of “any tank closure” vs “facility complete closure” by size bin. The ratio quantifies how much closure activity is partial vs total.

The “Any / Full” ratio rises sharply with size — at single-tank facilities the two rates are identical by definition; at 4+ tank facilities, “any tank closure” happens many times more often than “facility fully closes.” This is the portfolio-management signature: big facilities cycle individual tanks routinely without ever fully exiting.

5.7 How facility wall-state is assigned, and why DW-to-SW transitions exist

The DCM panel build (04b_Replacement_Panel_Prep.R, lines 117–119) classifies a facility-year’s `w_state` by the *worst tank’s wall type*:

```
w_state = DW    if wall_type == "Double-Walled"    (all tanks DW)
w_state = SW    otherwise                          (any SW tank, or unknown)
```

So a facility with 1 SW and 3 DW tanks has `wall_type = "Mixed"` at the facility level and is coded `w_state = SW` in the DCM panel. The intent (per the in-code comment) is: any SW tank in the facility’s portfolio means the facility still carries SW risk, so the SW classification dominates. This is the **worst-tank rule**.

Consequences:

1. **Mixed facilities are pooled into the SW cells.** A facility that’s retrofitted 3 of 4 tanks still shows up in SW state cells in the model. The hazard and premium cell averages in the SW pool are contaminated by these partially-retrofitted facilities.
2. **The 1 percent DW-to-SW transition rate observed earlier (Section 1) is a mechanical consequence of this rule, not a data error.** A facility coded “all DW” in year t that adds an SW tank in year $t + 1$ shifts to `wall_type = "Mixed"` and therefore to `w_state = SW`. The model’s transition matrix forbids this move (DW state never transitions to SW under any action), so these data points contradict the model’s deterministic wall evolution.
3. **Two alternative rules** are worth considering as robustness checks:
 - **Majority rule:** `w_state = DW` iff `pct_dw > 0.5`. Captures “what’s the typical tank at this facility?”
 - **All-DW rule:** `w_state = DW` iff `all_dw == 1` (every tank DW). Strictest interpretation – only facilities that have completed full retrofit count as DW. Symmetric to the current worst-tank rule.

The current worst-tank rule probably makes the SW pool noisier than necessary – it lumps “all SW” facilities together with “mostly DW with one straggler SW” facilities, even though these groups may have very different responses to retrofit incentives. A robustness check re-estimating the 6p+stayFE model under the majority rule (and reporting how $\hat{\theta}$ moves) would help characterize the sensitivity.

5.8 Partial closure as a candidate fourth action

Partial closure — a facility closes *some* tanks but keeps operating and never installs again — currently maps to **Maintain** under the T007 clean definitions (§4.5). This section characterizes partial-closure behavior across facility-portfolio dimensions and assesses whether it is large enough and structured enough to warrant adding as a fourth model action.

Table 9: Within-TX partial-closure rate by facility size and pricing regime (pre-1999 flat-fee vs 2006+ risk-based). Phase 0 diagnostic: T007. The bidirectional effect by size is the key finding — smaller facilities shrink less under RB, larger facilities shrink more.

Size bin	Regime	Closure events	N partial (orig)	Pct partial (orig)	Pct partial (clean)	Δ (clean—orig)
1	TX_FF_pre1999	12,655	0	0.0%	0.0%	+0.0pp
2	TX_FF_pre1999	9,723	909	9.3%	9.3%	+0.0pp
3	TX_FF_pre1999	7,451	1,073	14.4%	14.4%	+0.0pp
4+	TX_FF_pre1999	10,706	4,926	46.0%	46.0%	+0.0pp
1	TX_RB_2006plus	1,154	0	0.0%	0.0%	+0.0pp
2	TX_RB_2006plus	1,432	63	4.4%	4.4%	+0.0pp
3	TX_RB_2006plus	1,808	136	7.5%	7.5%	+0.0pp
4+	TX_RB_2006plus	2,173	1,134	52.2%	52.2%	+0.0pp

5.8.1 The bidirectional regime effect by facility size

The within-TX natural experiment (pre-1999 flat-fee vs 2006+ risk-based pricing) reveals a **bidirectional** partial-closure response that depends sharply on facility size (Table 9). Single-tank facilities show zero partial-closure events by construction (a single-tank closure is always a full exit). At **2–3 tank facilities**, the partial-closure rate is *lower* under RB pricing than under FF (−4.9pp to −6.9pp). At **4+ tank facilities**, the rate is *higher* under RB (+6.2pp). This sign reversal is the key finding.

The sign reversal survived a churn-tank filter: the 4+ tank effect is +6.2pp whether or not first-year-churn tanks are excluded (Phase 0 diagnostic result). Churn tanks are already filtered from the matched analysis panel by the `tstop > tstart` exposure criterion, so this bidirectional pattern is not a panel-build artifact.

Two structural-model design choices are motivated by this pattern:

1. **Add partial closure as a fourth action** — captures the response *level* (do facilities shrink at all?) but requires estimating a new cost parameter and extending the state space.
2. **Add facility size as a state dimension** — captures the *sign reversal* directly (size interacts with the regime-pricing effect) and may be sufficient if the level of partial shrinkage is well-approximated by allowing size-specific intercepts on the Maintain/Exit/Replace CCPs.

These two options are not mutually exclusive but have different implementation costs. Section Section 5.8.6 evaluates them together using the evidence from §§4.8.1–4.8.4.

5.8.2 Partial closure rate by facility-portfolio dimensions

Table 10 shows the unconditional partial-closure rate (fraction of closure events that are partial shrinkage, not full exit or retrofit) stratified along four portfolio dimensions: facility size, total capacity, wall composition, and average tank age.

Table 10: Partial-closure rate by four facility-portfolio dimensions, pooled across regime and year. Partial closure = any_closure = 1, facility_complete_closure = 0, no future install. **N closures** = denominator (any-closure events); **Pct partial** = fraction that are partial shrinkage.

Dimension	Level	N closures	N partial	Pct partial
Size bin	2	11,166	1,366	12.23
	1	12,631	0	0.00
	3	11,588	1,759	15.18
	4+	17,383	9,074	52.20
Capacity bin	<10K gal	18,038	1,513	8.39
	10-30K gal	20,839	3,966	19.03
	60K+ gal	3,185	2,498	78.43
	30-60K gal	10,432	4,218	40.43
Wall composition	All-SW	6,808	6,320	92.83
	All-DW	4,877	4,680	95.96
	Mixed	858	744	86.71
Avg tank age	10-20 yrs	17,120	5,444	31.80
	20+ yrs	31,849	5,148	16.16
	5-10 yrs	2,782	1,376	49.46
	<5 yrs	1,017	231	22.71

Table 11: Wall type of closed tanks at partial-closure, true-exit, and retrofit events. Rows: event type. Columns: fraction of closed-tank observations that are SW, DW, or Unknown wall. SW dominance at partial-closure events would support the risk-management interpretation.

Event type	N closed tanks	% SW	% DW
Partial shrinkage	26,466	90.0	10.0
Retrofit	21,513	89.3	10.7
True exit	83,637	90.8	9.2

5.8.3 Wall type of closed tank at partial-closure events

For each partial-closure event, Table 11 shows the wall type of the tank(s) that actually closed (identified by joining closure-year rows in `panel_dt`). The hypothesis: facilities preferentially close their oldest SW tanks (highest risk, least valuable), making partial closure a risk-management action.

5.8.4 Age of closed tank at partial-closure events

Figure 10 shows the distribution of tank age at closure for tanks involved in partial-closure events vs full exits and retrofits. The hypothesis: partial-closure tanks are older than average (firms pruning end-of-life tanks).

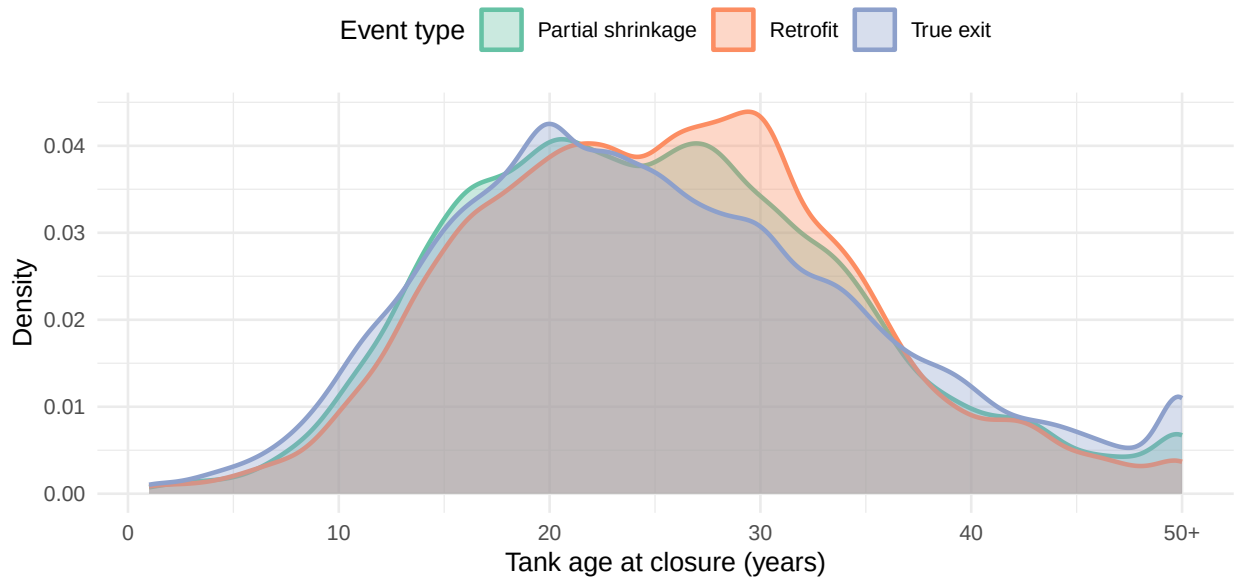
Table 12: Within-TX partial-closure rate by facility size and pricing regime, using T007 cleaned definitions. Phase 0 diagnostic: churn filter has zero effect on this sample (churn tanks already excluded by exposure filter). The 4+ tank bidirectional effect (+6.2pp) is the headline result.

Size	Regime	Pct (original)	Pct (clean)	Δ
1	TX_FF_pre1999	0.0%	0.0%	+0.0pp
2	TX_FF_pre1999	9.3%	9.3%	+0.0pp
3	TX_FF_pre1999	14.4%	14.4%	+0.0pp
4+	TX_FF_pre1999	46.0%	46.0%	+0.0pp
1	TX_RB_2006plus	0.0%	0.0%	+0.0pp
2	TX_RB_2006plus	4.4%	4.4%	+0.0pp
3	TX_RB_2006plus	7.5%	7.5%	+0.0pp
4+	TX_RB_2006plus	52.2%	52.2%	+0.0pp

Figure 10: Distribution of tank age at closure (years since installation) by event type. Partial-shrinkage tanks, true-exit tanks, and retrofit tanks. Partial-closure tanks systematically older than retrofit tanks would support the risk-management interpretation.

Age of closed tank by event type

Partial-shrinkage tanks compared to true-exit and retrofit tanks



5.8.5 Policy response check: within-TX 1999 cutoff

Table 12 repeats the within-TX partial-closure test using the post-T007 cleaned action definitions and confirms the bidirectional finding from Phase 0. Since churn tanks are already excluded from the matched analysis panel (by the `tstop > tstart` filter in 02b), the cleaned and original rates are identical in this sample; the column is shown for transparency.

5.8.6 Recommendation: 4th action or size as state dimension?

The four analyses above converge on a specific characterization of partial closure: it is concentrated at large multi-tank facilities (4+ tanks), is associated with SW tanks (the higher-risk wall type), and involves disproportionately old tanks. The within-TX regime test shows a **+6.2pp increase in partial-closure frequency at 4+ tank facilities under RB pricing**, which is a meaningful behavioral response to the pricing regime.

Two model extensions are motivated by this evidence, and they answer different questions:

Option A — Add partial closure as a 4th model action. This directly represents the partial-shrinkage choice in the value function. It requires (i) defining the fourth action at the facility-year level in 04b, (ii) adding a new partial-closure cost parameter to the utility specification, (iii) extending the estimator and state-transition matrix to handle four actions, and (iv) re-running the full estimation suite. The benefit: the welfare CF (Phase 4) would capture the social cost of suppressed or induced partial closure under alternative pricing regimes. The cost: substantial implementation work, plus the fourth action is only empirically relevant at 4+ tank facilities (25% of the sample), so it may not move the aggregate welfare number enough to justify the effort.

Option B — Add facility size as a state dimension. Rather than adding an action, extend the state space from 32 to 128 cells (4 size bins $\times$ existing 32). This captures the bidirectional size-regime interaction directly in the CCPs without a new action parameter. The implementation is lighter — the existing 6p+stayFE+profile machinery supports arbitrary additive group intercepts, and size FE are already the specification tested in the T004 profile-likelihood architecture. The cost: the partial-closure behavior would still be folded into the Maintain action, and the welfare CF cannot distinguish “facilities shrinking their portfolio” from “facilities simply not exiting or replacing.”

Recommendation: implement Option B (size as state dimension) first. The evidence for a size-differentiated structural response is strong (6.2pp regime effect at 4+ tank facilities, sign reversal relative to 2–3 tank facilities, consistent wall-type and age patterns). Option B can be implemented as a modest extension to the existing 6p+stayFE+profile spec and will immediately improve the model’s empirical fit. If, after adding size FE, the partial-closure-specific CF contribution to welfare remains large (a separate computation), Option A can be added in a subsequent ticket with the size-differentiated baseline already in place. Option A without Option B would add a new action but still average over size heterogeneity — and the Phase 0 evidence suggests the behavioral margin is precisely where size matters most.

6 Implications

6.1 For the dynamic structural model

6.1.1 Action definitions (highest priority)

Section 5.5 shows the most consequential model-data mismatch: the “Partial shrinkage” row — facility closed some tanks, kept operating, and never installs again — is currently classified as **Exit** under T005 but is behaviorally **Maintain** under the user’s clean facility-level definition. Fixing this requires only a small change to 04b’s action assignment, then re-running the T005 estimation suite on the corrected panel. The expected result: $\hat{\kappa}$ moves (partial-shrinkage signal is removed from the Exit margin), $\hat{K}$ may shift modestly, and the structural welfare counterfactuals become defensible. **This fix should land before any CF estimation.**

6.1.2 Size and capacity (second priority)

The 6-parameter universal- γ specification (T005) treats every facility as if it were a single representative tank with the cell-average (P, h) values. The size and capacity diagnostics above show that:

- **Single-tank facilities (23\% of sample) and 4+ tank facilities (25\%) have qualitatively different action profiles**, especially at the DW Replace margin where 4+ tank facilities replace at $\sim 4\%$ vs $\sim 0\%$ for single-tank.
- The “Replace” action in the model conflates two distinct firm behaviors: structural retrofit (SW to DW upgrade) and tank-level portfolio refresh at multi-tank facilities.
- Capacity adds incremental information beyond tank count.

The cleanest minimal extension is to add **facility size as a profile-FE grouping dimension** (analogous to state FE), interacted with state or additive. This requires no new estimator architecture — the existing T004 profile-likelihood machinery generalizes to any number of additive group intercepts.

Recommended ticket: **add size FE (4 bins: 1, 2, 3, 4+) to the 6p+stayFE+profile specification**, either additive (3 new profiled-out intercepts) or interacted with state FE (17 times 4 = 68 intercepts). Compare to the T005 fit on LL improvement. If

Delta

logL is large, this is a real source of misspecification; if small, the model is robust to size and the diagnostic can be left as a footnote.

6.2 For the reduced-form analyses

The event-study and DiD specifications elsewhere in the project currently aggregate at the facility-year level without explicit size controls. The size patterns above suggest:

- **Stratify event-study estimates by facility size** (e.g.,

Table 13: Action coding: 02b flags mapped to 04b DCM actions

[!h]		
Category	N	Pct
Maintain	2238133	98.0461157
Exit	38392	1.6818422
Replace	6210	0.2720421
PartialClosure_inMaintain	8134	0.3563269
CompleteExit_noFutureInstall	38392	1.6818422
CompleteExit_withFutureInstall	1310	0.0573873

leq2 tanks vs

geq3 tanks). The headline ATT may differ across size classes; reform effects could plausibly be larger at large facilities (which have more decisions to make per year).

- **Add size as a control variable** in OLS / Cox specifications. At a minimum include `n_tanks_active` (or its log) as a covariate.
- **Capacity may add additional explanatory power** — worth testing whether `log(total_capacity)` improves fit beyond size.
- **Report compositional shifts** — if the treated and control samples differ systematically in size composition, the matching / weighting procedures should be re-examined.

6.3 For paper exposition

Make the per-tank-vs-per-facility distinction explicit:

- All structural cost estimates (*kappa*, *K*) are **per-tank**
- All flow utility components (revenue, premium, hazard loss) are **per-tank**
- Per-facility welfare effects scale with average tanks-per-facility (*approx2.7* in this sample)
- The agent in the model is best understood as “a representative tank at a facility making one closure-or-replace decision per year,” acknowledging that 25\% of observed closure events involve only some tanks at multi-tank facilities (which the model treats as composite full-facility events)

7 T011 Diagnostics: action coding, empirical CCPs, and size variation

7.1 Action coding audit

7.1.1 Action distribution: 02b flags vs 04b `y_it`

Notes: Six categories computed on the merged estimation sample; the common denominator is all sample rows, so the categories overlap by design (a partial closure is also a Maintain). Partial closures (`any_closure = 1`, `facility_complete_closure = 0`) are classified Maintain in the DCM. `CompleteExit_withFutureInstall` is the

Table 14: Closure-type overlap: complete closure $\times$ replacement flag

[!h]			
Complete closure	Replacement flag	N	Pct of closures
1	1	1310	2.937088
1	0	38392	86.076858
0	1	4900	10.986054

double-triggered case in which a full closure coincides with a future install, which the model codes as Replace (retrofit wins).

7.1.2 Closure type overlap: full-exit $\times$ replacement flag

Notes: Cross-tabulation among closure rows ($y_it = 1$). Pct of closures uses total estimation-sample closures as the denominator. Rows with $facility_complete_closure = 1$ and $replacement_closure_year = 1$ are the double-triggered events mapped to Replace.

7.1.3 Replacement margin validity: timing, tank count, and capacity change

Table 15: Replacement-margin validity: timing, tank count, and capacity change

Section	Category	N	Pct of Replace events	
Timing (years to next install)	0	16371	61.2297565	b
Timing (years to next install)	1	3957	14.7997157	b
Timing (years to next install)	2-3	2136	7.9889292	b
Timing (years to next install)	4-5	1166	4.3609979	b
Timing (years to next install)	6-10	1596	5.9692561	b
Timing (years to next install)	11+	1246	4.6602087	b
Timing (years to next install)	NA	265	0.9911359	b
Net tank count change (t-1 to t+1)	+2 or more	821	3.0706512	b
Net tank count change (t-1 to t+1)	+1	1924	7.1960205	b
Net tank count change (t-1 to t+1)	0	7959	29.7677376	b
Net tank count change (t-1 to t+1)	-1	7128	26.6596851	b
Net tank count change (t-1 to t+1)	-2 or more	8151	30.4858436	b
Net capacity change pct (t-1 to t+1)	expanded >10pct	10916	40.8273179	b
Net capacity change pct (t-1 to t+1)	flat -10 to +10pct	5890	22.0293975	b
Net capacity change pct (t-1 to t+1)	modest shrink -25 to -10pct	2883	10.7828103	b
Net capacity change pct (t-1 to t+1)	substantial shrink below -25pct	6294	23.5404122	b
Net capacity change pct (t-1 to t+1)	missing capacity data	754	2.8200621	b
SW-replacement install timing (SW closure + future install)	same year	14446	61.5037466	b
SW-replacement install timing (SW closure + future install)	next year	3510	14.9438011	b
SW-replacement install timing (SW closure + future install)	2 or more years later	5330	22.6924387	b

Notes: Replace events are facility-years with $replacement_closure_year = 1$ in the full facility panel. Before/after stock is anchored by calendar year ($t-1$ and the next-install year) via a gap-safe join. The tank-count and capacity sub-blocks exclude events with a missing before/after neighbor row. SW-replacement timing uses the broader set of

Table 16: Replace-definition concordance: broad, SW-to-DW, and DCM coding

[!h]				
Category	N	Pct of DCM Replace	Pct of all closures	Note
DCM_Replace_total	6210	100.00000	13.923143	all DCM Replace events (I_replace=
DCM_Replace_is_SW2DW	2870	46.21578	6.434689	DCM Replace that are same-year SW
DCM_Replace_broad_not_SW2DW	3340	53.78422	7.488453	DCM Replace that are broad future-
broad_replace_NOT_in_DCM_replace	0	0.00000	0.000000	broad replacement flag but NOT cod
SW2DW_NOT_in_DCM_replace	0	0.00000	0.000000	SW->DW flag but NOT coded DCM
Identity_check_discrepancies	0	0.00000	0.000000	rows where I_replace != replacement

Table 17: Partial closures by facility size

[!h]					
Size bin	N partial	N obs (size)	N closures (size)	Pct obs partial	Pct closures partial
1	470	529950	12965	0.0886876	3.625145
2	2042	581238	10417	0.3513191	19.602573
3	3721	614875	11524	0.6051637	32.289136
4+	6800	555805	9670	1.2234507	70.320579

facility-years with an SW closure and a future install; the same-year SW→DW count (single_to_double_year = 1) is reported in the note column for cross-check.

7.1.4 Replace definition: broad vs SW-to-DW vs DCM coding

Notes: Three replace definitions exist in the pipeline: broad (replacement_closure_year, any future install; equals 04b is_retrofit), SW→DW (single_to_double_year, an SW tank closed and a DW tank installed the same year), and DCM (equals broad). Rows 4–6 are validity checks that should all equal zero: SW→DW is a subset of broad, and 04b is_retrofit reproduces 02b replacement_closure_year exactly.

7.1.5 Partial closure breakdown by facility size

Notes: Partial-closure events (any_closure = 1, facility_complete_closure = 0) in the estimation sample, binned by beginning-of-year facility size (prior-year end-of-year tank stock; see Section 1.5). Rows with size bin “unknown” are excluded. Partial closures are always Maintain in the DCM.

7.1.6 Distribution of size change at Replace events

Table 18: Distribution of facility size change at replace events

Metric	Bin	N events	Pct of events
delta_tanks	<=-5	863	3.3214025
delta_tanks	-4	812	3.1251203
delta_tanks	-3	2073	7.9782935
delta_tanks	-2	4403	16.9456953
delta_tanks	-1	7128	27.4333218
delta_tanks	0	7959	30.6315668
delta_tanks	+1	1924	7.4048416
delta_tanks	+2	568	2.1860447

(continued)

Metric	Bin	N events	Pct of events
delta_tanks	+3	170	0.6542739
delta_tanks	+4	61	0.2347689
delta_tanks	$\geq +5$	22	0.0846707
delta_cap_gal	≤ -10000	4377	16.8456298
delta_cap_gal	-10000 to -5000	2486	9.5677943
delta_cap_gal	-5000 to -2000	2572	9.8987800
delta_cap_gal	-2000 to -500	1357	5.2226456
delta_cap_gal	-500 to <0	241	0.9275295
delta_cap_gal	0 (no change)	2490	9.5831890
delta_cap_gal	>0 to 500	503	1.9358812
delta_cap_gal	500 to 2000	2090	8.0437209
delta_cap_gal	2000 to 5000	2455	9.4484855
delta_cap_gal	5000 to 10000	3285	12.6428819
delta_cap_gal	>10000	4127	15.8834623

Notes: Distribution of facility size change at Replace events (replace-event population with non-missing before/after stock). Tank count is in integer bins clipped to $[-5, +5]$; capacity is in gallon bins with an explicit no-change bin, extending to $\pm 10,000$ gallons (about the p10–p90 range). Pct of events sums to 100 within each metric block. Capacity is reported in gallons rather than percent because tank sizes are standardized in the thousands of gallons, which makes percent change heavy-tailed.

7.1.7 Capacity-change distribution figure

Figure 11: Distribution of facility size change at replace events

Distribution of facility size change at Replace events

Tank count (top) and total capacity in gallons (bottom); $n = 25983$ replace events with non-missing before/after stock. Mass at/above 0 on capacity =

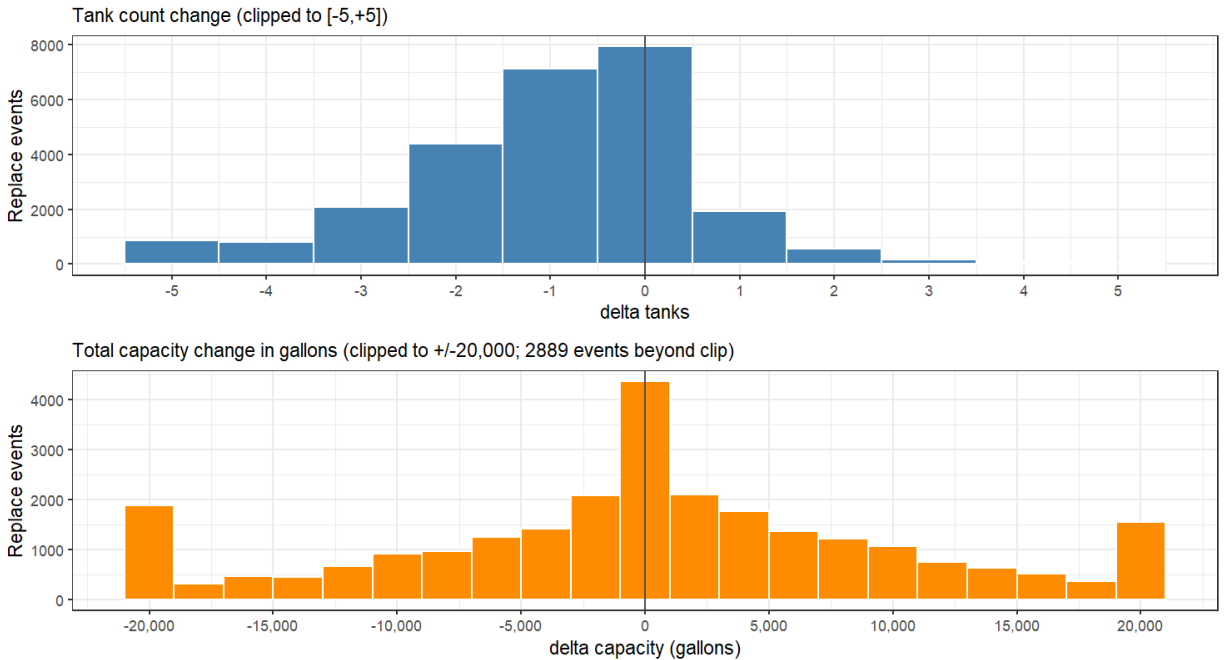


Table 19: Tank-count change versus capacity change at replace events

[!h]							
Tank chg bin	N	Pct	Mean cap pct	Median cap pct	P25 cap pct	P75 cap pct	Pct cap-preserving
<=-5	863	3.3214025	418.5309	-48.27586	-73.79863	-19.56702	16.68598
-4	812	3.1251203	6615.0646	-39.71983	-71.39024	-10.36391	24.87685
-3	2073	7.9782935	5526.7424	-26.82927	-52.38095	-1.35635	32.94742
-2	4403	16.9456953	3323.1624	-14.57859	-40.00000	13.69022	45.78696
-1	7128	27.4333218	7164.9911	0.00000	-25.00000	33.03769	61.64422
0	7959	30.6315668	5383.8774	10.70111	0.00000	54.54545	84.57093
+1	1924	7.4048416	18929.9222	64.44612	25.00000	150.00000	95.21830
+2	568	2.1860447	33859.8252	132.57576	53.51190	353.01522	97.53521
+3	170	0.6542739	42317.1617	237.50000	88.91076	937.50000	99.41176
+4	61	0.2347689	61864.9945	336.95652	136.36364	950.00000	96.72131
>=+5	22	0.0846707	2297.6946	527.14617	61.66047	2625.00000	100.00000

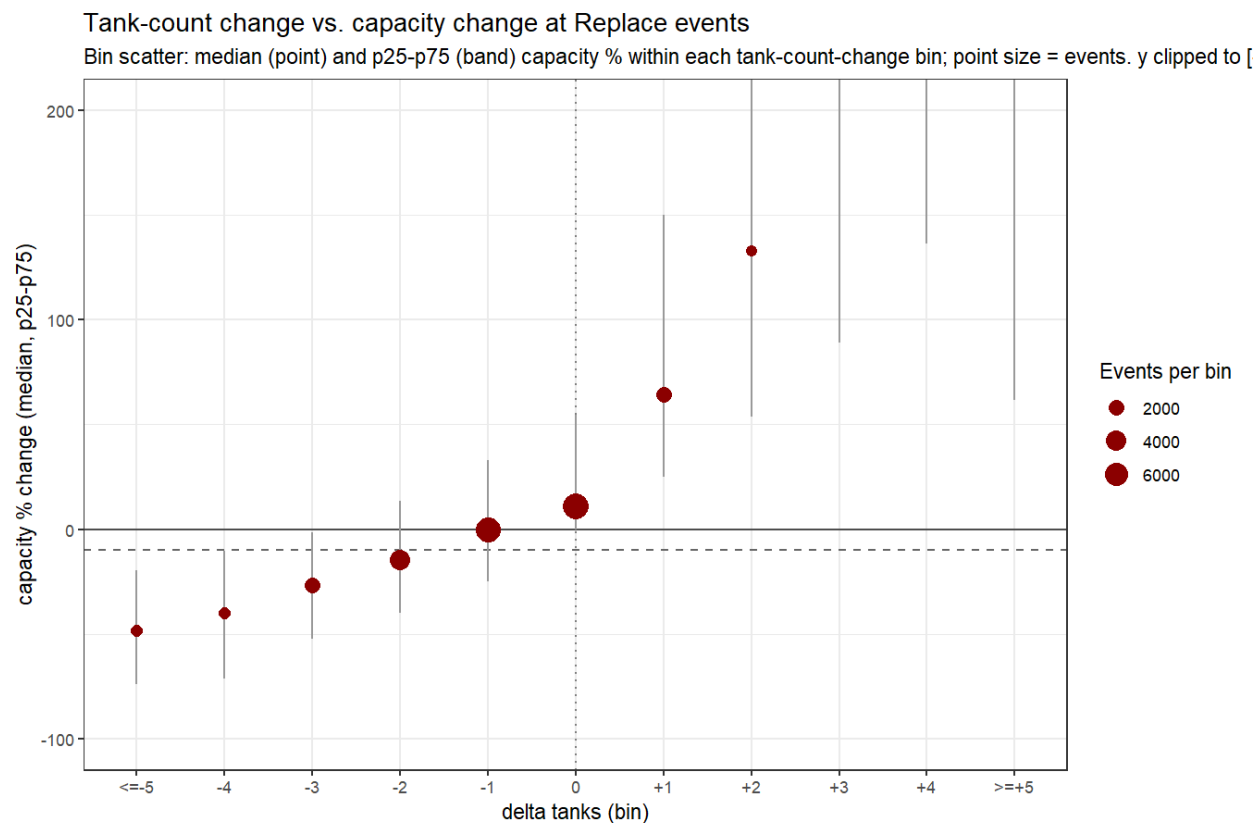
Notes: Two-panel histogram of facility size change at Replace events: tank count (top, clipped to $[-5, +5]$) and total capacity in gallons (bottom, clipped to $\pm 20,000$; events beyond the clip are noted in the panel subtitle). Vertical line marks no change. The capacity distribution is roughly symmetric around zero, consistent with consolidation rather than systematic shrinkage.

7.1.8 Tank-count vs capacity change (joint)

Notes: Joint distribution of tank-count and capacity change at Replace events. For each tank-count-change bin: median and interquartile range of capacity percent change, and the share that is capacity-preserving ($\text{delta_cap_pct} \geq -10$). A negative tank-count bin with median capacity near zero is consolidation (fewer, larger tanks); a negative bin with strongly negative capacity is true shrinkage. The mean column is dominated by small-denominator outliers and is reported only for completeness; the figure plots the median.

7.1.9 Tank-count vs capacity change bin scatter

Figure 12: Tank-count change versus capacity change at replace events (bin scatter)



Notes: Bin scatter of capacity percent change against tank-count change at Replace events: median (point) and p25-p75 (band) within each tank-count-change bin; point area is the number of events. The y-axis is clipped to $[-100, +200]$ so the bands and reference lines are visible. Solid line at zero (capacity flat); dashed line at -10 (shrink threshold); dotted vertical line at zero tank change. Near zero capacity at negative tank change is consolidation; well below zero is true facility shrinkage.

7.2 Empirical CCPs by state cell

7.2.1 Raw empirical CCP table — all 32 cells

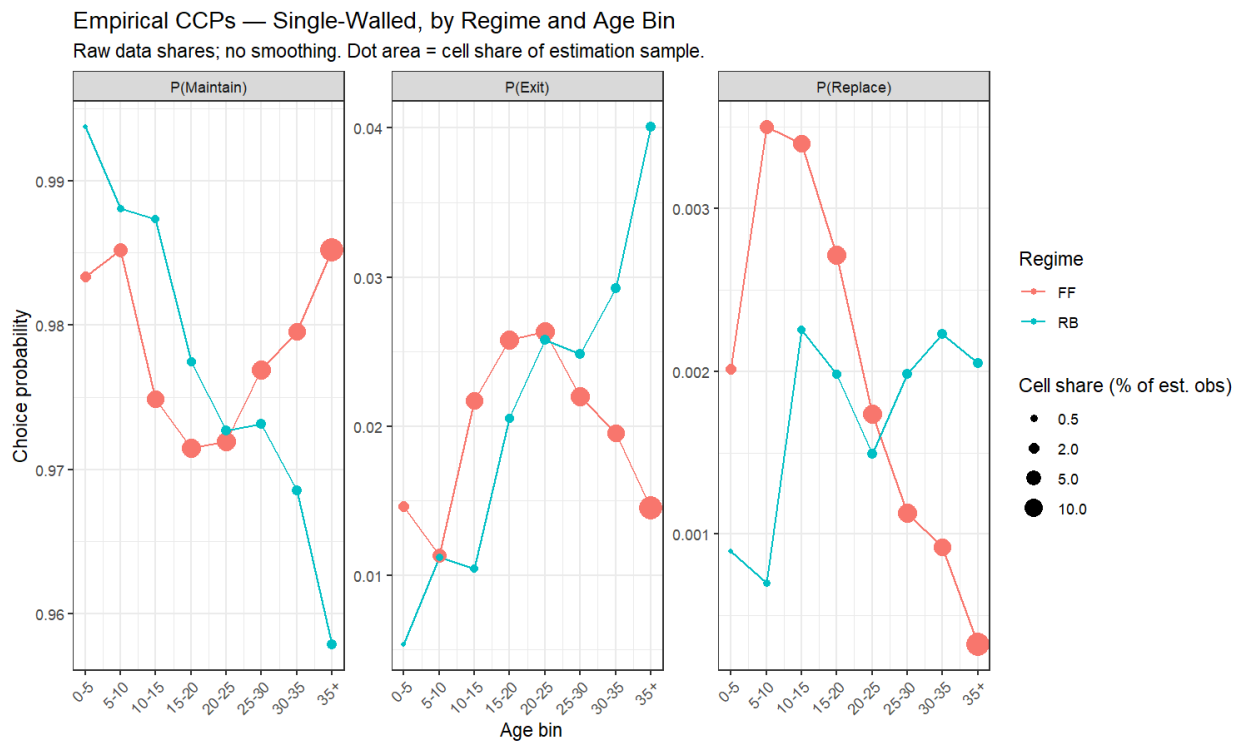
Notes: Raw empirical choice probabilities by state cell (age $\times$ wall $\times$ regime; 32 cells). Wall: SW = single-walled, DW = double-walled. Regime: FF = flat-fee, RB = risk-based. $P(M)+P(E)+P(R)=1$ by construction; $P(E|cl)$ and $P(R|cl)$ are conditional on a closure and sum to 1. Every DW closure is coded Replace, so $P(R|cl)=1$ for all DW cells. Cell pct is the cell's share of the estimation sample; sparse cells should be read with caution.

Table 20: Empirical choice probabilities by state cell

[!h]										
s	Age	Wall	Regime	N obs	P(M)	P(E)	P(R)	P(E cl)	P(R cl)	Cell pct
1	0-5	SW	FF	46709	0.9833	0.01464	0.00201	0.879	0.121	2.05
2	5-10	SW	FF	97999	0.9852	0.01131	0.00350	0.764	0.236	4.29
3	10-15	SW	FF	180163	0.9749	0.02174	0.00340	0.865	0.135	7.89
4	15-20	SW	FF	240344	0.9715	0.02580	0.00271	0.905	0.095	10.53
5	20-25	SW	FF	255278	0.9719	0.02634	0.00174	0.938	0.062	11.18
6	25-30	SW	FF	234030	0.9769	0.02198	0.00113	0.951	0.049	10.25
7	30-35	SW	FF	191377	0.9795	0.01953	0.00092	0.955	0.045	8.38
8	35+	SW	FF	410728	0.9852	0.01450	0.00032	0.978	0.022	17.99
9	0-5	DW	FF	64621	0.9990	0.00000	0.00099	0.000	1.000	2.83
10	5-10	DW	FF	77103	0.9936	0.00000	0.00636	0.000	1.000	3.38
11	10-15	DW	FF	77966	0.9908	0.00000	0.00920	0.000	1.000	3.42
12	15-20	DW	FF	62241	0.9893	0.00000	0.01067	0.000	1.000	2.73
13	20-25	DW	FF	41520	0.9931	0.00000	0.00686	0.000	1.000	1.82
14	25-30	DW	FF	20407	0.9974	0.00000	0.00265	0.000	1.000	0.89
15	30-35	DW	FF	4872	0.9947	0.00000	0.00534	0.000	1.000	0.21
16	35+	DW	FF	2867	0.9962	0.00000	0.00384	0.000	1.000	0.13
17	0-5	SW	RB	3350	0.9937	0.00537	0.00090	0.857	0.143	0.15
18	5-10	SW	RB	9999	0.9881	0.01120	0.00070	0.941	0.059	0.44
19	10-15	SW	RB	19093	0.9873	0.01042	0.00225	0.822	0.178	0.84
20	15-20	SW	RB	26212	0.9775	0.02056	0.00198	0.912	0.088	1.15
21	20-25	SW	RB	36841	0.9727	0.02581	0.00149	0.945	0.055	1.61
22	25-30	SW	RB	37789	0.9732	0.02485	0.00198	0.926	0.074	1.66
23	30-35	SW	RB	31888	0.9685	0.02923	0.00223	0.929	0.071	1.40
24	35+	SW	RB	30722	0.9579	0.04007	0.00205	0.951	0.049	1.35
25	0-5	DW	RB	18813	0.9995	0.00000	0.00053	0.000	1.000	0.82
26	5-10	DW	RB	16557	0.9943	0.00000	0.00568	0.000	1.000	0.73
27	10-15	DW	RB	14499	0.9926	0.00000	0.00738	0.000	1.000	0.64
28	15-20	DW	RB	13106	0.9802	0.00000	0.01976	0.000	1.000	0.57
29	20-25	DW	RB	8677	0.9749	0.00000	0.02512	0.000	1.000	0.38
30	25-30	DW	RB	4218	0.9829	0.00000	0.01707	0.000	1.000	0.18
31	30-35	DW	RB	1770	0.9808	0.00000	0.01921	0.000	1.000	0.08
32	35+	DW	RB	976	0.9816	0.00000	0.01844	0.000	1.000	0.04

7.2.2 Empirical CCPs — single-walled panel

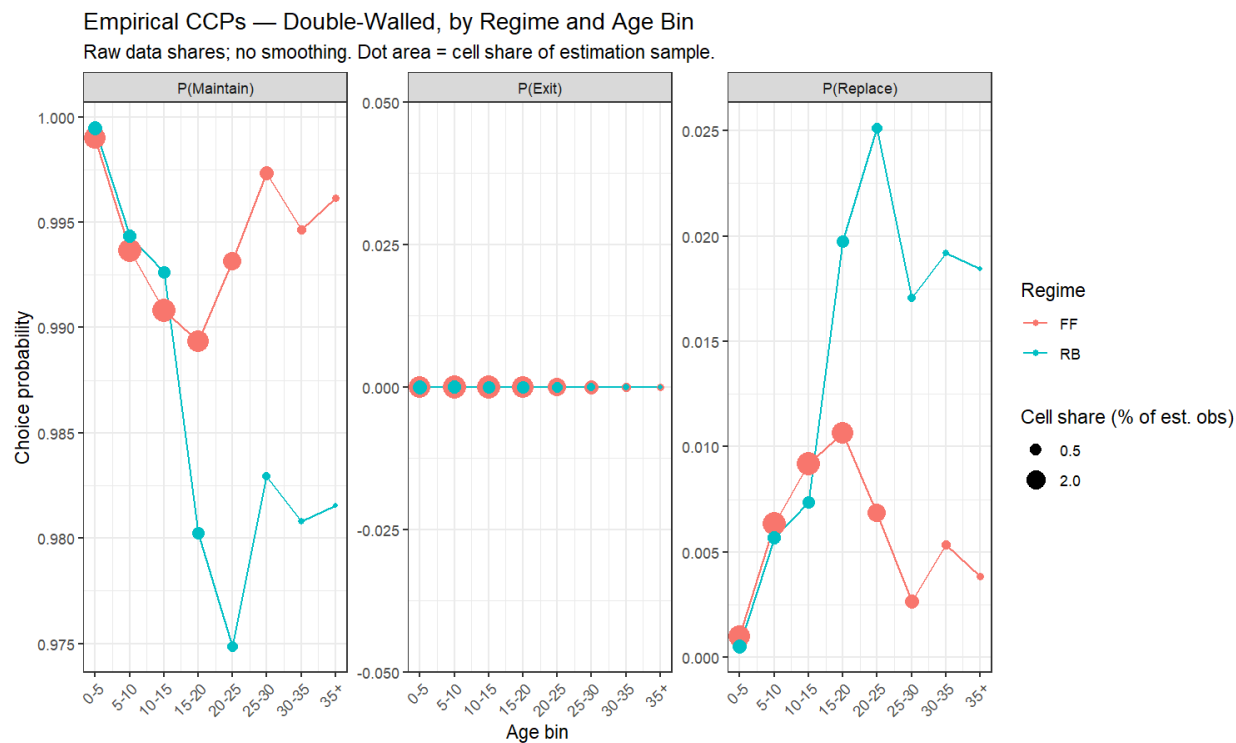
Figure 13: Empirical choice probabilities, single-walled facilities



Notes: Single-walled facilities. Raw data shares by age bin, faceted by action (Maintain, Exit, Replace); no smoothing. Dot area is the cell's share of the estimation sample. FF = flat-fee, RB = risk-based.

7.2.3 Empirical CCPs — double-walled panel

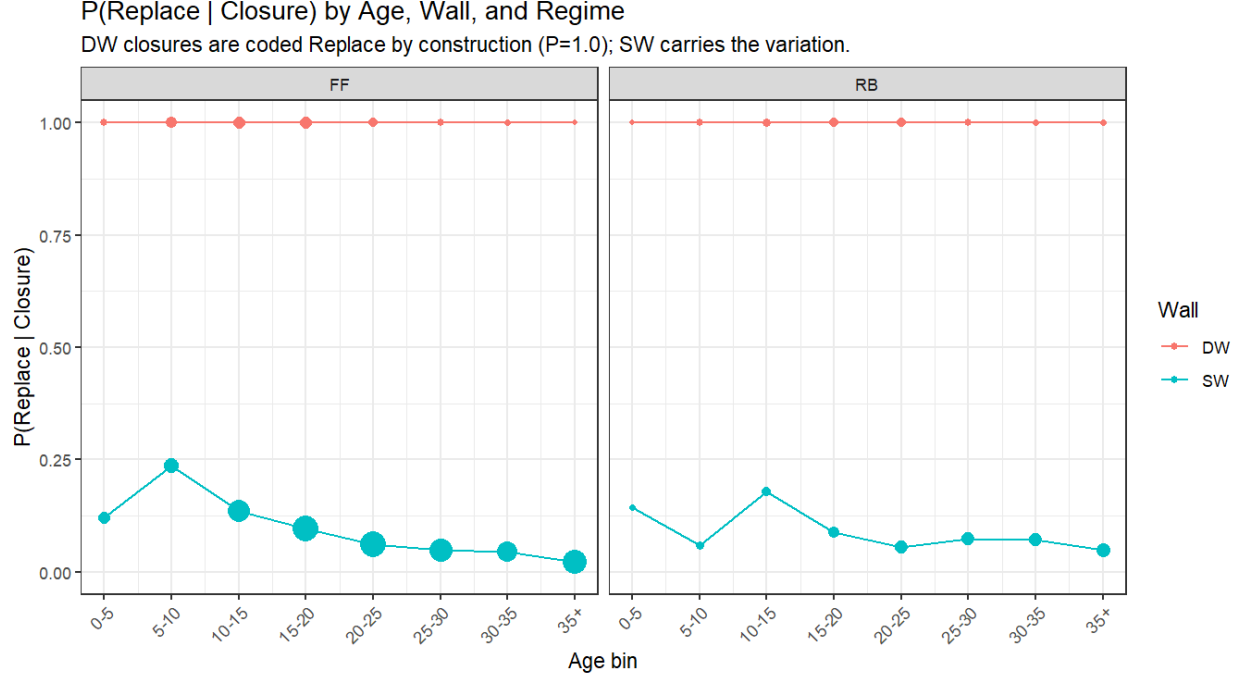
Figure 14: Empirical choice probabilities, double-walled facilities



Notes: Double-walled facilities. Raw data shares by age bin, faceted by action; no smoothing. Dot area is the cell's share of the estimation sample. Double-walled closures are coded Replace by construction, so the Exit panel is empty. FF = flat-fee, RB = risk-based.

7.2.4 P(Replace | Closure) by age, wall, and regime

Figure 15: Replace share among closures, by age, wall, and regime



Notes: Conditional replace share among closures, faceted by pricing regime (FF, RB) and colored by wall type. Double-walled closures are coded Replace by construction (P = 1.0 in every cell), so the single-walled series carries all the identifying variation. Point area is the number of closures in the cell.

7.3 CCP variation by facility size

7.3.1 CCPs by (size bin x state cell)

Table 21: Empirical choice probabilities by facility size and state cell

Size	Age	Wall	Regime	N obs	P(M)	P(E)	P(R)	P(E cl)	P(R cl)
1	0-5	SW	FF	9914	0.9572	0.03904	0.00373	0.913	0.087
1	5-10	SW	FF	19280	0.9702	0.02619	0.00363	0.878	0.122
1	10-15	SW	FF	33079	0.9487	0.04973	0.00160	0.969	0.031
1	15-20	SW	FF	43487	0.9506	0.04854	0.00090	0.982	0.018
1	20-25	SW	FF	48906	0.9530	0.04678	0.00022	0.995	0.005
1	25-30	SW	FF	49233	0.9687	0.03110	0.00016	0.995	0.005
1	30-35	SW	FF	46427	0.9783	0.02158	0.00015	0.993	0.007
1	35+	SW	FF	146549	0.9868	0.01317	0.00004	0.997	0.003
1	0-5	DW	FF	12009	0.9984	0.00000	0.00158	0.000	1.000
1	5-10	DW	FF	17907	0.9953	0.00000	0.00469	0.000	1.000
1	10-15	DW	FF	20331	0.9978	0.00000	0.00221	0.000	1.000
1	15-20	DW	FF	17177	0.9990	0.00000	0.00105	0.000	1.000
1	20-25	DW	FF	12403	0.9999	0.00000	0.00008	0.000	1.000

(continued)

Size	Age	Wall	Regime	N obs	P(M)	P(E)	P(R)	P(E cl)	P(R cl)
1	25-30	DW	FF	6584	1.0000	0.00000	0.00000	0.000	0.000
1	30-35	DW	FF	1468	1.0000	0.00000	0.00000	0.000	0.000
1	35+	DW	FF	1083	1.0000	0.00000	0.00000	0.000	0.000
1	0-5	SW	RB	1113	0.9919	0.00629	0.00180	0.778	0.222
1	5-10	SW	RB	3059	0.9853	0.01471	0.00000	1.000	0.000
1	10-15	SW	RB	5029	0.9879	0.01133	0.00080	0.934	0.066
1	15-20	SW	RB	5123	0.9588	0.04060	0.00059	0.986	0.014
1	20-25	SW	RB	4360	0.9365	0.06353	0.00000	1.000	0.000
1	25-30	SW	RB	2896	0.9275	0.07251	0.00000	1.000	0.000
1	30-35	SW	RB	2183	0.9290	0.07055	0.00046	0.994	0.006
1	35+	SW	RB	2597	0.9322	0.06700	0.00077	0.989	0.011
1	0-5	DW	RB	4786	0.9990	0.00000	0.00104	0.000	1.000
1	5-10	DW	RB	3769	0.9966	0.00000	0.00345	0.000	1.000
1	10-15	DW	RB	3278	0.9991	0.00000	0.00092	0.000	1.000
1	15-20	DW	RB	2942	0.9993	0.00000	0.00068	0.000	1.000
1	20-25	DW	RB	1861	0.9995	0.00000	0.00054	0.000	1.000
1	25-30	DW	RB	797	1.0000	0.00000	0.00000	0.000	0.000
1	30-35	DW	RB	242	1.0000	0.00000	0.00000	0.000	0.000
1	35+	DW	RB	78	1.0000	0.00000	0.00000	0.000	0.000
2	0-5	SW	FF	16757	0.9879	0.01068	0.00143	0.882	0.118
2	5-10	SW	FF	31727	0.9877	0.00987	0.00246	0.801	0.199
2	10-15	SW	FF	45930	0.9742	0.02297	0.00285	0.890	0.110
2	15-20	SW	FF	53940	0.9694	0.02937	0.00128	0.958	0.042
2	20-25	SW	FF	53894	0.9695	0.02954	0.00095	0.969	0.031
2	25-30	SW	FF	46106	0.9754	0.02410	0.00048	0.981	0.019
2	30-35	SW	FF	38146	0.9787	0.02087	0.00039	0.982	0.018
2	35+	SW	FF	97096	0.9838	0.01597	0.00022	0.987	0.013
2	0-5	DW	FF	23515	0.9991	0.00000	0.00085	0.000	1.000
2	5-10	DW	FF	25952	0.9964	0.00000	0.00362	0.000	1.000
2	10-15	DW	FF	23585	0.9958	0.00000	0.00424	0.000	1.000
2	15-20	DW	FF	17922	0.9967	0.00000	0.00335	0.000	1.000
2	20-25	DW	FF	10741	0.9980	0.00000	0.00196	0.000	1.000
2	25-30	DW	FF	4352	0.9986	0.00000	0.00138	0.000	1.000
2	30-35	DW	FF	951	0.9989	0.00000	0.00105	0.000	1.000
2	35+	DW	FF	548	0.9964	0.00000	0.00365	0.000	1.000
2	0-5	SW	RB	1884	0.9942	0.00531	0.00053	0.909	0.091
2	5-10	SW	RB	5426	0.9891	0.01032	0.00055	0.949	0.051
2	10-15	SW	RB	9565	0.9895	0.01004	0.00042	0.960	0.040
2	15-20	SW	RB	10207	0.9823	0.01656	0.00118	0.934	0.066
2	20-25	SW	RB	8980	0.9748	0.02416	0.00100	0.960	0.040
2	25-30	SW	RB	6680	0.9651	0.03353	0.00135	0.961	0.039
2	30-35	SW	RB	5552	0.9557	0.04359	0.00072	0.984	0.016
2	35+	SW	RB	6884	0.9542	0.04503	0.00073	0.984	0.016
2	0-5	DW	RB	9926	0.9997	0.00000	0.00030	0.000	1.000
2	5-10	DW	RB	8611	0.9931	0.00000	0.00685	0.000	1.000
2	10-15	DW	RB	6826	0.9963	0.00000	0.00366	0.000	1.000
2	15-20	DW	RB	5368	0.9933	0.00000	0.00671	0.000	1.000
2	20-25	DW	RB	2763	0.9949	0.00000	0.00507	0.000	1.000
2	25-30	DW	RB	900	0.9900	0.00000	0.01000	0.000	1.000

(continued)

Size	Age	Wall	Regime	N obs	P(M)	P(E)	P(R)	P(E cl)	P(R cl)
2	30-35	DW	RB	300	0.9900	0.00000	0.01000	0.000	1.000
2	35+	DW	RB	204	0.9951	0.00000	0.00490	0.000	1.000
3	0-5	SW	FF	11875	0.9936	0.00514	0.00126	0.803	0.197
3	5-10	SW	FF	26390	0.9892	0.00765	0.00311	0.711	0.289
3	10-15	SW	FF	50962	0.9826	0.01430	0.00306	0.824	0.176
3	15-20	SW	FF	68760	0.9765	0.02065	0.00282	0.880	0.120
3	20-25	SW	FF	73388	0.9773	0.02122	0.00153	0.933	0.067
3	25-30	SW	FF	66403	0.9780	0.02084	0.00111	0.949	0.051
3	30-35	SW	FF	51189	0.9773	0.02163	0.00109	0.952	0.048
3	35+	SW	FF	80688	0.9838	0.01574	0.00051	0.969	0.031
3	0-5	DW	FF	18194	0.9992	0.00000	0.00077	0.000	1.000
3	5-10	DW	FF	21200	0.9915	0.00000	0.00854	0.000	1.000
3	10-15	DW	FF	20762	0.9858	0.00000	0.01421	0.000	1.000
3	15-20	DW	FF	16356	0.9806	0.00000	0.01938	0.000	1.000
3	20-25	DW	FF	10729	0.9897	0.00000	0.01025	0.000	1.000
3	25-30	DW	FF	5256	0.9952	0.00000	0.00476	0.000	1.000
3	30-35	DW	FF	1307	0.9939	0.00000	0.00612	0.000	1.000
3	35+	DW	FF	511	0.9980	0.00000	0.00196	0.000	1.000
3	0-5	SW	RB	261	0.9962	0.00383	0.00000	1.000	0.000
3	5-10	SW	RB	1147	0.9887	0.00785	0.00349	0.692	0.308
3	10-15	SW	RB	3342	0.9832	0.01137	0.00539	0.679	0.321
3	15-20	SW	RB	7333	0.9802	0.01677	0.00300	0.848	0.152
3	20-25	SW	RB	14372	0.9787	0.01969	0.00160	0.925	0.075
3	25-30	SW	RB	17048	0.9791	0.01918	0.00176	0.916	0.084
3	30-35	SW	RB	14768	0.9772	0.02038	0.00237	0.896	0.104
3	35+	SW	RB	13688	0.9630	0.03404	0.00292	0.921	0.079
3	0-5	DW	RB	2611	0.9996	0.00000	0.00038	0.000	1.000
3	5-10	DW	RB	3101	0.9932	0.00000	0.00677	0.000	1.000
3	10-15	DW	RB	3442	0.9855	0.00000	0.01453	0.000	1.000
3	15-20	DW	RB	3740	0.9607	0.00000	0.03930	0.000	1.000
3	20-25	DW	RB	2998	0.9613	0.00000	0.03869	0.000	1.000
3	25-30	DW	RB	1705	0.9789	0.00000	0.02111	0.000	1.000
3	30-35	DW	RB	848	0.9800	0.00000	0.02005	0.000	1.000
3	35+	DW	RB	501	0.9900	0.00000	0.00998	0.000	1.000
4+	0-5	SW	FF	7887	0.9938	0.00431	0.00190	0.694	0.306
4+	5-10	SW	FF	20602	0.9902	0.00427	0.00548	0.438	0.562
4+	10-15	SW	FF	50192	0.9849	0.00970	0.00542	0.642	0.358
4+	15-20	SW	FF	74157	0.9806	0.01466	0.00472	0.756	0.244
4+	20-25	SW	FF	79090	0.9803	0.01627	0.00341	0.827	0.173
4+	25-30	SW	FF	72288	0.9823	0.01548	0.00221	0.875	0.125
4+	30-35	SW	FF	55615	0.9833	0.01498	0.00176	0.895	0.105
4+	35+	SW	FF	86395	0.9853	0.01392	0.00075	0.949	0.051
4+	0-5	DW	FF	10438	0.9989	0.00000	0.00105	0.000	1.000
4+	5-10	DW	FF	12044	0.9891	0.00000	0.01088	0.000	1.000
4+	10-15	DW	FF	13288	0.9792	0.00000	0.02085	0.000	1.000
4+	15-20	DW	FF	10786	0.9751	0.00000	0.02494	0.000	1.000
4+	20-25	DW	FF	7647	0.9800	0.00000	0.02001	0.000	1.000
4+	25-30	DW	FF	4215	0.9945	0.00000	0.00546	0.000	1.000
4+	30-35	DW	FF	1146	0.9852	0.00000	0.01483	0.000	1.000

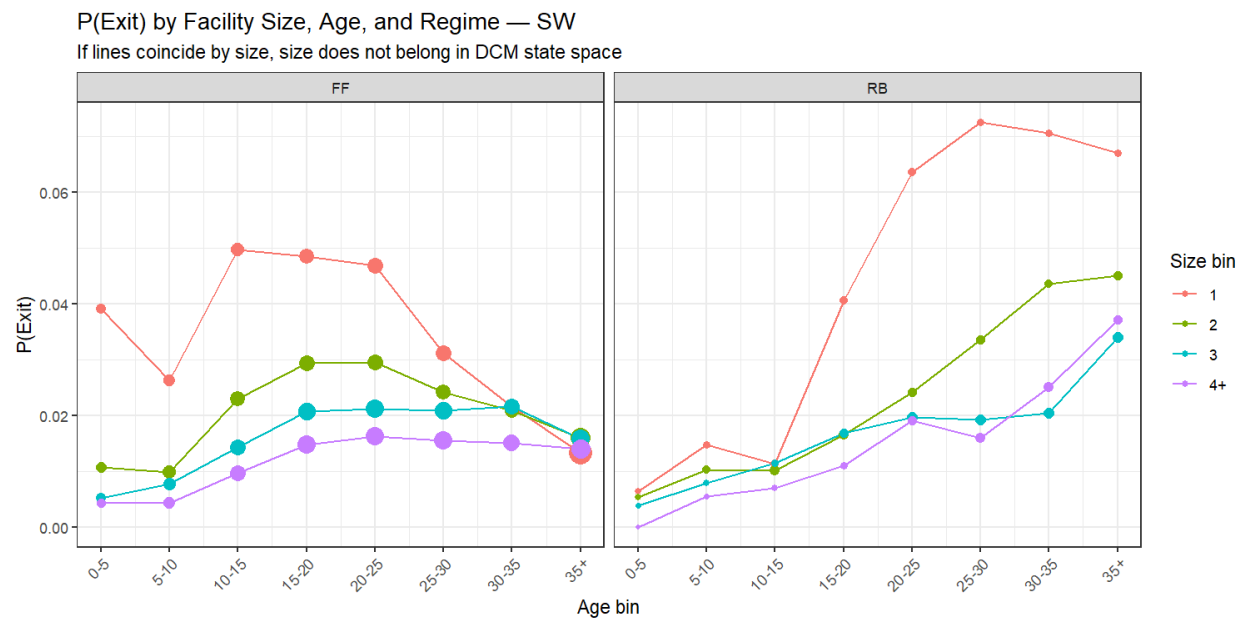
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Size	Age	Wall	Regime	N obs	P(M)	P(E)	P(R)	P(E cl)	P(R cl)
4+	35+	DW	FF	725	0.9890	0.00000	0.01103	0.000	1.000
4+	0-5	SW	RB	80	1.0000	0.00000	0.00000	0.000	0.000
4+	5-10	SW	RB	367	0.9946	0.00545	0.00000	1.000	0.000
4+	10-15	SW	RB	1157	0.9784	0.00691	0.01469	0.320	0.680
4+	15-20	SW	RB	3549	0.9848	0.01099	0.00423	0.722	0.278
4+	20-25	SW	RB	9129	0.9784	0.01906	0.00252	0.883	0.117
4+	25-30	SW	RB	11165	0.9808	0.01594	0.00322	0.832	0.168
4+	30-35	SW	RB	9385	0.9717	0.02504	0.00330	0.883	0.117
4+	35+	SW	RB	7553	0.9607	0.03720	0.00212	0.946	0.054
4+	0-5	DW	RB	1376	0.9993	0.00000	0.00073	0.000	1.000
4+	5-10	DW	RB	1076	0.9991	0.00000	0.00093	0.000	1.000
4+	10-15	DW	RB	953	0.9696	0.00000	0.03043	0.000	1.000
4+	15-20	DW	RB	1056	0.9299	0.00000	0.07008	0.000	1.000
4+	20-25	DW	RB	1055	0.9175	0.00000	0.08246	0.000	1.000
4+	25-30	DW	RB	816	0.9669	0.00000	0.03309	0.000	1.000
4+	30-35	DW	RB	380	0.9632	0.00000	0.03684	0.000	1.000
4+	35+	DW	RB	193	0.9378	0.00000	0.06218	0.000	1.000
unknown	0-5	SW	FF	276	0.9058	0.08333	0.01087	0.885	0.115
unknown	0-5	DW	FF	465	1.0000	0.00000	0.00000	0.000	0.000
unknown	0-5	DW	RB	114	1.0000	0.00000	0.00000	0.000	0.000

Notes: Empirical CCPs by (size bin $\times$ state cell); cells with fewer than 20 observations excluded. Facility size is the beginning-of-year tank count (Section 1.5). $P(E|cl)$ and $P(R|cl)$ are conditional on a closure. If the conditional shares are roughly constant across size bins within the same (age, wall, regime) cell, facility size does not belong in the DCM state space.

7.3.2 Exit rate by size x age (SW)

Figure 16: Exit rate by facility size and age (single-walled)



Notes: Exit rate by facility size, age, and regime, single-walled facilities only; cells with fewer than 20 observations dropped. Four lines correspond to the four beginning-of-year size bins. Coincident lines imply facility size does not belong in the DCM state space.

7.3.3 Replace rate by size x age (SW)

Figure 17: Replace rate by facility size and age (single-walled)

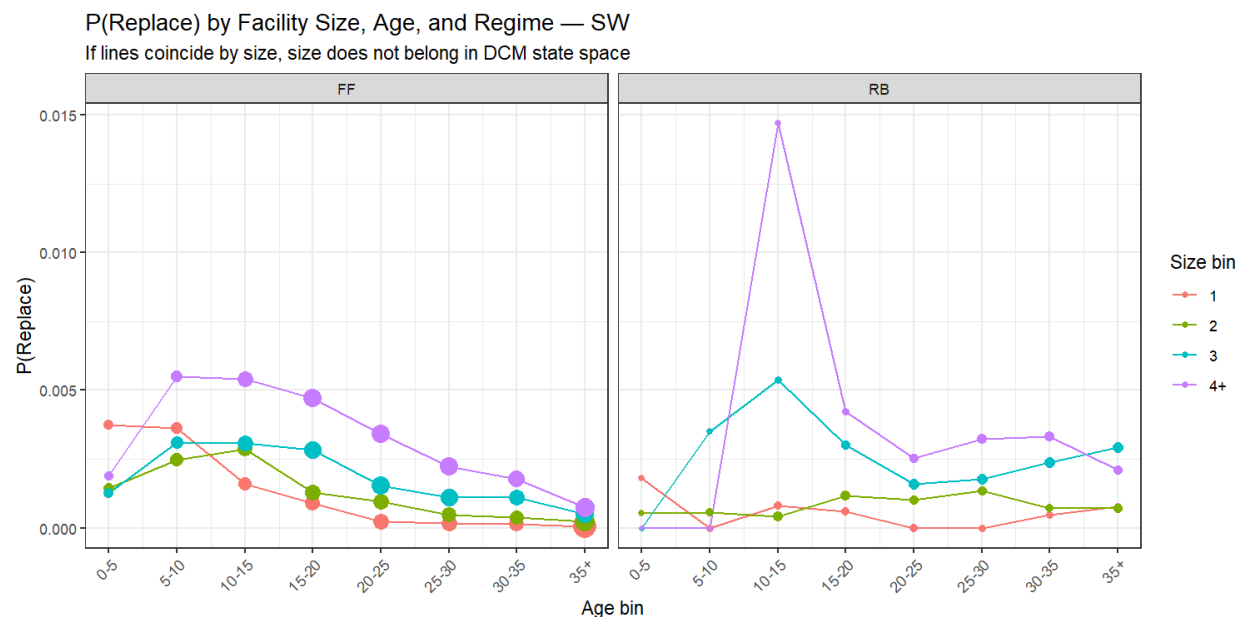


Table 22: Closure composition by facility size and pricing regime

[!h]								
Size	Regime	N events	N partial	N full exit	N full replace	Pct partial	Pct exit	Pct replace
1	Control_FF	55774	2739	52189	846	4.9	93.6	1.5
1	TX_FF	13039	375	12514	150	2.9	96.0	1.2
1	TX_RB	2453	56	2377	20	2.3	96.9	0.8
2	Control_FF	42909	9029	33023	857	21.0	77.0	2.0
2	TX_FF	9998	1239	8579	180	12.4	85.8	1.8
2	TX_RB	2825	378	2379	68	13.4	84.2	2.4
3	Control_FF	35826	11810	22880	1136	33.0	63.9	3.2
3	TX_FF	8203	1946	5918	339	23.7	72.1	4.1
3	TX_RB	4117	993	2882	242	24.1	70.0	5.9
4+	Control_FF	47737	27157	19180	1400	56.9	40.2	2.9
4+	TX_FF	9276	4310	4614	352	46.5	49.7	3.8
4+	TX_RB	3036	1101	1793	142	36.3	59.1	4.7

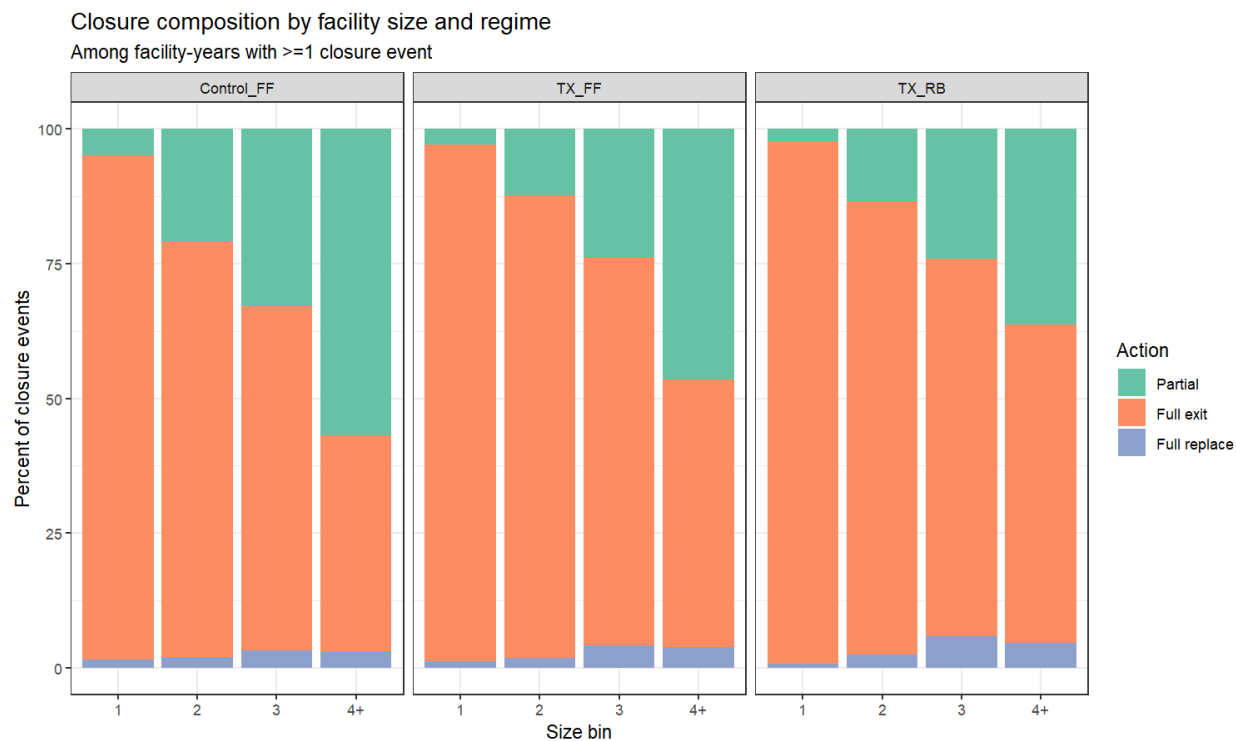
Notes: Replace rate by facility size, age, and regime, single-walled facilities only; cells with fewer than 20 observations dropped. Four lines correspond to the four beginning-of-year size bins. Coincident lines imply facility size does not belong in the DCM state space.

7.3.4 Partial closure composition by size and regime

Notes: Closure composition among facility-years with at least one closure, by beginning-of-year size bin and pricing regime, from the full facility panel (not restricted to the DCM estimation sample). Pct partial = portfolio shrinkage (some tanks closed, facility survives); pct full exit = complete walk-away with no future install; pct full replace = complete closure with a future install. The three shares sum to 100 within each row.

7.3.5 Closure composition stacked bar

Figure 18: Closure composition by facility size and regime



Notes: Closure composition (partial, full exit, full replace) by beginning-of-year facility size and regime, among facility-years with at least one closure. Bars are shown only for size $\times$ regime cells with at least 30 closure events. Partial closures rise steeply with facility size.